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Review

Affective Computing for Learning in Education: A Systematic Review and Bibliometric Analysis

Rajamanickam Yuvaraj [1,*,†], Rakshit Mittal [2,†], A. Amalin Prince [3] and Jun Song Huang [1]

1. Science of Learning in Education Center (SoLEC), Office of Education Research (OER), National Institute of Education (NIE), Nanyang Technological University (NTU), 1 Nanyang Walk, Singapore 637616, Singapore
2. Department of Computer Science, University of Antwerp, FlandersMake @ UAntwerpen, 2020 Antwerp, Belgium
3. Department of Electrical and Electronics Engineering, BITS Pilani, K K Birla Goa Campus, Sancoale 403726, Goa, India

* Correspondence: yuvaraj.rajamanickam@nie.edu.sg or yuva2257@gmail.com
† These authors contributed equally to this work.

Abstract: Affective computing is an emerging area of education research and has the potential to enhance educational outcomes. Despite the growing number of literature studies, there are still deficiencies and gaps in the domain of affective computing in education. In this study, we systematically review affective computing in the education domain. Methods: We queried four well-known research databases, namely the Web of Science Core Collection, IEEE Xplore, ACM Digital Library, and PubMed, using specific keywords for papers published between January 2010 and July 2023. Various relevant data items are extracted and classified based on a set of 15 extensive research questions. Following the PRISMA 2020 guidelines, a total of 175 studies were selected and reviewed in this work from among 3102 articles screened. The data show an increasing trend in publications within this domain. The most common research purpose involves designing emotion recognition/expression systems. Conventional textual questionnaires remain the most popular channels for affective measurement. Classrooms are identified as the primary research environments; the largest research sample group is university students. Learning domains are mainly associated with science, technology, engineering, and mathematics (STEM) courses. The bibliometric analysis reveals that most publications are affiliated with the USA. The studies are primarily published in journals, with the majority appearing in the *Frontiers in Psychology* journal. Research gaps, challenges, and potential directions for future research are explored. This review synthesizes current knowledge regarding the application of affective computing in the education sector. This knowledge is useful for future directions to help educational researchers, policymakers, and practitioners deploy affective computing technology to broaden educational practices.

Keywords: affective computing; learning; education; computerized teaching; emotion recognition; emotion regulation; literature review; PRISMA

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Academic Editor: Diego Vergara

Received: 8 November 2024
Revised: 26 December 2024
Accepted: 27 December 2024
Published: 10 January 2025

Citation: Yuvaraj, R., Mittal, R., Prince, A. A., & Huang, J. S. (2025). Affective Computing for Learning in Education: A Systematic Review and Bibliometric Analysis. *Education Sciences*, *15*(1), 65. https://doi.org/10.3390/educsci15010065

1. Introduction

Emotions are strongly linked to human behavior, family, and society. They influence a person's psychological and behavioral activities, such as processing information, perception, thinking ability, and decision-making (Damasio, 1995). Naturally, as computers developed and more systems were automated, it became important to integrate the emotions of users within a computer's decision or interface, making them more suitable for use by humans.

Rosalind Picard's book 'Affective Computing' was published in 1997, and it offered the first glimpse into the answer to solving this need. Since then, affective computing has evolved considerably to fill this gap by integrating emotions with human–computer interaction (HCI). In the context of affective computing, emotional interactions between a human and a computer are provided by recognizing the user's emotional state and then regulating the user's emotion through emotion-aware reinforcements provided by the computer.

In Pekrun (2000, 2006), the control-value theory (CVT) of achievement emotions was described, which has laid the groundwork for much of the theory behind the emotion detection and regulation systems that are being implemented today. The CVT argues that student emotions are tied to their sense of satisfaction and achievement in the learning environment. This means students' emotions can be changed by shifting the subjective control and value of achievement activities to foster a more positive learning psychology. The CVT also goes on to show that positive valence and arousal can lead to better student performance and learning outcomes. Valence is the pleasantness/unpleasantness of the emotions, whereas arousal is the intensity of that emotion. Many other studies have since built on this work, for example, Hoffman (2009) and Peterson et al. (2015) demonstrated that emotions significantly impact student learning and academic performance. Studies (Lackéus, 2014) have shown a significant relationship between students' emotional states and the development of entrepreneurial skills. In educational environments, affective computing can be employed to build emotion-aware systems and enhance user experiences (Handayani et al., 2014). In other words, affective computing can play an essential role in the field of education/learning since knowledge acquisition can be significantly improved by engaging learners emotionally. Emotional information can provide insights to enhance learning (Linnenbrink-Garcia et al., 2016) through various emotional regulation interventions.

Thus, the integration of affective computing capabilities can significantly advance the frontiers of educational technologies by enhancing the overall learning outcomes (Caballé, 2015). Furthermore, having an enhanced emotional state has several benefits for learners, such as promoting higher cognitive flexibility (Malekzadeh et al., 2015). Several research studies have explored different aspects of affective computing in education. Many studies (Baldassarri et al., 2013) have designed and developed emotion awareness models and platforms for educational environments namely affective tutoring systems (ATSs), and intelligent tutoring systems (ITSs) as demonstrated by der Kleij et al. (2015).

From 2020 to 2021, learners all around the world were forced to shift to computer-based learning methods. This was a significant cognitive and emotional load for both students and teachers. In such a scenario, affective computing in education is an apt solution to mitigate increased emotional loads. Research in this field was bolstered, and an increased number of publications in the domain was witnessed. Since this field is still relatively young and highly scattered due to many different complex models of emotions and systems, it is essential to summarize the published literature to complement future research.

In this regard, this article systematically reviews the affective computing field in the context of education to understand the current developments and trends. Education is composed of two parallel processes: teaching and learning. While the former focuses on teachers' well-being, which is correlated with their teaching performance, the impact on student achievement is indirect. Hence, in this article, the authors specifically focus on literature that deals with learning-related emotions in the educational context. Systematically pre-defined research questions aim to comprehensively review the articles in the field by identifying and summarizing various characteristics of the varied articles, including research trends, learning domains, affective measurements, learning environments, theoretical models of emotions, research sample groups, and emotional states. Furthermore, based on bibliometric analyses, this paper summarizes the influential articles, highest contributing countries, publication venues, and research topics in affective computing for education. The relationships and links among the publications and citations are also mapped for complementary information.

The rest of this article is constructed as follows. The 15 research questions formulated for this study are presented. Then, the review methodology employed in this study is described. The related work, i.e., previous literature reviews, are presented. Next, the results of the systematic review based on the set of formulated research questions are reported. The outcomes, challenges, and potential directions for future research are discussed. The conclusions are described. The Supplementary Materials present the data extracted during the review, from the reviewed articles.

2. Current Study Objectives

As described before, affective computing is an exponentially growing domain, with various new developments in a concise span of time. The need to collate all the information from these developments into a single review will be very beneficial for future research. A timely high-level review of applications of affective computing in education can help readers/researchers identify gaps in research in this domain. A review of this domain will also help developers by providing a summary of the published state-of-the-art. To conduct a formal analysis of the literature, we define various research questions that cover different aspects of applications of affective computing in education. These questions are listed systematically in Table 1.

Table 1. Research questions in this study.

RQ No.	Research Question
	Systematic Review
RQ1.	Which learning domains does the research focus on?
RQ2.	What affective measurements are used?
RQ3.	What datasets are used/available?
RQ4.	What types of affective systems are proposed?
RQ5.	What kinds of examinations are conducted for validation?
RQ5.1	What are the environments in these examinations?
RQ5.2	What are the sample sizes in these examinations?
RQ5.3	What are the sample groups in these examinations?
RQ6.	How well do the interventions perform?
RQ7.	What theoretical models of emotion are reported and used?
RQ8.	What emotions are reported?
	Additional Data Analysis
RQ9.	What are the temporal trends?
RQ10.	Which are the venues for the publication of research?
RQ11.	What is the quality of work published?
RQ12.	What is the research purpose of the published work?
	Bibliometric Review
RQ13.	How are the publications bibliometrically connected?
RQ14.	Which countries publish work in the domain?
RQ15.	What are the schools of thought?

This study identifies 15 research questions (RQ1 to RQ15). Some of these questions are composed of multiple sub-questions. The research questions cover a wide range of characteristics of the domain, i.e., affective computing research to complement learning in education.

A systematic review employs rigorous and well-defined methods to extract data from the literature in order to answer specific research questions within any given scientific domain. In our review, we have employed the PRISMA methodology (Page et al., 2021). PRISMA is a standardized methodology for literature reviews that involves precise definition of research questions and design of a search strategy (involving querying and filtering) based on the research question. The review methodology for the current review is precisely defined in Section 3. In our review, research questions RQ1–RQ8 have been formulated to systematically analyze the domain. RQ5 discusses explicitly the various kinds of experiments and their characteristics.

The term meta-data refers to data about primary data. In our case, primary data consist of the articles selected for this review. Primary data include the publication year, quality score, research purpose category, etc., giving us vital statistical information on the overall composition of the data. A statistical analysis of the meta-data in our review is facilitated through research questions RQ9 through RQ12.

In a bibliometric analysis, research in a particular domain is measured through unique indicators like affiliations, citations, and co-citations to identify characteristics like the most cited researchers, affiliated countries, frequently used keywords, etc. In our review, questions RQ13–RQ15 cover the bibliometric analysis. Questions regarding the existence of schools of thought, collaboration between researchers/organizations, and maturity of the domain are answered through bibliometric indicators studied in the corresponding sections.

3. Review Methodology Based on PRISMA

The systematic review process typically begins by preparing a review protocol based on the underlying research questions and methods predefined by Kitchenham and Charters (2007). According to Balaid et al. (2016), the main components of a review protocol include reviewing the background of the subject, formulating the research questions, developing a research strategy, selection criteria, and a data extraction strategy, and, finally, synthesis of the data obtained. The research background and questions of this study are indicated in previous sections. The review protocol of this study is illustrated in Figure 1.

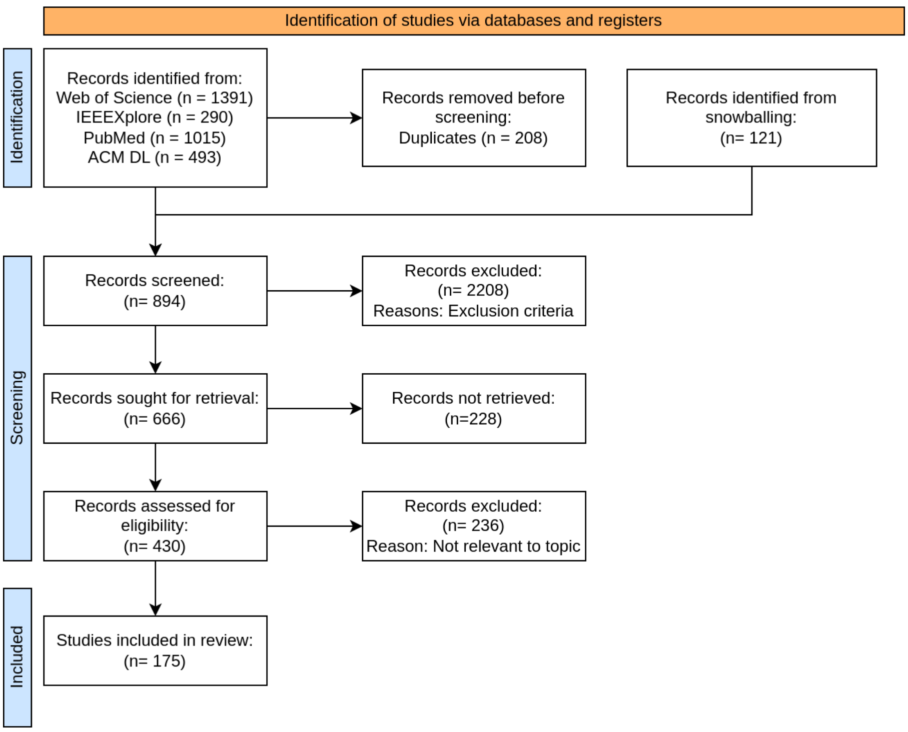

Figure 1. The review protocol (following PRISMA 2020 guidelines flow diagram) used in this study.

3.1. PRISMA-Based Review Protocol

In this study, the authors followed the preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines (Page et al., 2021) to conduct the review. The PRISMA guidelines specify the steps to be reported for the identification of documents used in a systematic review.

3.1.1. Search Strategy

The first step, according to the PRISMA guidelines, is the formulation of the search strategy. The authors choose five well-known bibliometric databases to collect the review data: Science Citation Index Expanded (Clarivate Analytics, n.d.-a), Social Science Citation Index (Clarivate Analytics, n.d.-b), IEEE Xplore (Institute of Electrical and Electronics Engineers, n.d.), ACM Digital Library (Association for Computing Machinery, n.d.), and PubMed National Library of Medicine (n.d.).

1. The Science Citation Index and the Social Science Citation Index are parts of a larger collection, namely the Web of Science (WoS) Core Collection offered and maintained by Clarivate Analytics (https://clarivate.com/academia-government/scientific-and-academic-research/research-discovery-and-referencing/web-of-science/web-of-science-core-collection/ (accessed on 26 December 2024)). This is a collection of 21,100 of the world's highest-impact journals, proceedings, and book chapters from 1900 to the present. It contains over 74.8 million records dating back to 1900.
2. PubMed is a bibliometric database consisting of more than 7 million records spanning several biomedical and life science research domains. It is hosted and maintained by the US National Library of Medicine.
3. IEEEXplore is a bibliometric service provided by the Institute of Electrical and Electronics Engineers (IEEE), which is a collection of over 5.4 million articles from electronics, computer science, and allied fields.
4. ACM Digital Library is a bibliometric service provided by the Association for Computing Machinery providing access to more than 0.8 million full-text articles published since 1951 from the computer scientific domain.

3.1.2. Search Query

To make the review broad enough to capture all aspects of affective computing in education, while leaving out unrelated literature, the authors formulated a search query as follows: ((“affective computing” OR “emotion” OR “emotion recognition”) AND (“teaching” OR “student” OR “teacher” OR “education”)).

1. The first sub-query, i.e., “affective computing” OR “emotion” OR “emotion recognition”, searches for terms related to affective computing.
2. The second sub-query, i.e., “teaching” OR “student” OR “teacher” OR “education”, defines terms related to education. While the term “learning" is related to education, it is not included to avoid false positives associated with just “machine learning” and not “education”.

3.1.3. Criteria for Inclusion and Exclusion

To narrow down the data obtained from the search strategy, various inclusion and exclusion criteria were defined. The inclusion criteria defined grounds for papers to be included in the review, while the exclusion criteria defined characteristics that were unacceptable. The inclusion criteria are as follows:

1. The paper is written in English. English is the predominant language of scientific literature, ensuring that the widest audience can access and engage with the content.

This inclusion criterion reflects the practical necessity of limiting the language scope to those publications accessible to the majority of researchers in the field.

2. The paper has undergone a standard independent review process before being published. Peer-reviewed papers provide a higher level of credibility and scientific rigor, ensuring that the findings and conclusions are based on sound research methods and validated by experts in the field. This criterion ensures the inclusion of reliable, scientifically rigorous studies.
3. The paper is an original contribution or a review. Original contributions present new findings, whereas reviews synthesize and analyze existing literature. Both types of papers are crucial for advancing knowledge in the field, which is why they are prioritized for inclusion in this review.
4. The paper is published in the period 2010–2023. The most recently published reviews of affective computing in education consider literature from 2018–2022 (Lek & Teo, 2023) and 2016 (Khadimallah et al., 2020). For reasons described before, there has been a considerable increase in research in this domain in recent years, hence, the authors decided to include the entire literature from 2010 to 2023 in the scope of this article.

The exclusion criteria are as follows:

1. The paper is an opinion piece like editorials, newspaper articles, or gray literature. Opinion pieces are typically not subject to peer review and often lack the scientific rigor required for academic research. Such materials are excluded because they do not contribute novel, evidence-based findings or validated hypotheses to the scientific community.
2. The paper does not contribute to answering the review questions i.e. the paper is not within the scope of this review. The present review focuses on affective computing-based methods that support learning in an educational context. Papers that discuss other aspects, such as teachers' emotions or general pedagogical theories, while important, do not directly address the review's primary focus and are, therefore, excluded.
3. The paper does not give sufficient proof to validate the hypotheses. For inclusion, the paper must present robust evidence supporting its hypotheses, whether through experimental data, thorough analysis, or well-supported theoretical arguments. Papers lacking sufficient validation weaken the conclusions and would not meet the standards for this review.
4. The paper is a position paper/short paper. Position papers and short papers often present arguments or opinions without sufficient empirical evidence or comprehensive analysis. These types of publications are excluded because they tend to lack the depth and rigorous analysis required for inclusion in a systematic review of scientific literature. Usually, such papers are typically accompanied by full-length papers on the same subject, offering more rigorous findings (which are included in the review if they are found in the initial corpus).

3.2. Study Selection

The studies shortlisted through the inclusion/exclusion criteria needed to be manually verified to confirm that they were within the scope of this review. To do so, the studies went through three rounds of selection carried out by two independent researchers. The first round consisted of a selection of studies based on their titles. The second round was a more in-depth analysis that included text within each study's abstract. The third round was used to fine-tune the selection by an analysis of the entire content of each study.

After these three rounds were conducted independently by the two researchers, their selections were compared and conflicts were resolved through discussion. The study selection process was designed this way to ensure the least amount of bias.

3.3. Data Extraction

The authors queried the four bibliographic databases, using the defined search strategy, to obtain the initial pool of articles, in August 2023. A total of 1391 articles were retrieved from the Web of Science Core Collection. The search query yielded 1015 articles from PubMed, 493 articles from ACM Digital Library, and 290 articles from the IEEE Xplore database. This variation can be attributed to several factors, such as the scope of each database, the specific query terms used, and the different indexing practices across databases. PubMed, for example, focuses on biomedical and life sciences literature, which may lead to a higher yield of relevant articles. In contrast, the ACM Digital Library and IEEE Xplore focus more on computer science and engineering fields, which might result in a lower number of articles, but potentially with a greater emphasis on computational aspects of affective computing in education.

The initial search criteria were applied uniformly across all databases, but the inherent differences in their content and indexing methodologies naturally led to disparities in article counts. The pool of articles was unified and overlaps were eliminated to result in a final pool of 3102 articles.

At this stage, in this pool, the authors also included all the articles from past literature reviews of affective computing in education, to ensure they did not miss any literature. The pool of articles was then filtered using inclusion and exclusion criteria. A total of 894 articles were finally obtained, which matched the inclusion and exclusion criteria.

The pool of 894 articles was then independently reviewed by the two reviewers through three rounds of increasing scrutiny, as described before. Finally, after resolving the conflicts between the reviewers, the final pool of articles reviewed in this paper stands at exactly 175 articles. The entire methodology is described using PRISMA-style notation in Figure 1.

Various data pertaining to the research questions were extracted manually from the 175 selected articles. These data include the following:

1. First affiliation country (see Section 7.2).
2. Availability (open or closed-source) (see Section 6.2).
3. Venue type (see Section 6.2).
4. Year (see Section 6.1).
5. Quality score (see Section 6.3).
6. Research purpose category (see Section 6.4).
7. Experiment environment (see Section 5.5.1).
8. Proposed intervention type (see Section 5.4).
9. Validation methodology (see Section 5.5).
10. Unique approach (see Section 7.3).
11. Target learning domain (see Section 5.1).
12. Sample group type (see Section 5.5.3).
13. Sample size (no. of students in study) and age distribution (see Section 5.5.2).
14. Emotion model used in the study (see Section 5.7).
15. Considered emotions (see Section 5.8).
16. Results (see Section 5.6).
17. Used datasets (see Section 5.3).
18. Used channels for affective measurement and their types (see Section 5.2).

Papers that presented relevant datasets were subject to the following data extraction parameters:

1. First affiliation country (see Section 7.2).
2. Availability (open- or closed-source) (see Section 6.2).

3. Venue type (see Section 6.2).
4. Year (see Section 6.1).
5. Quality score (see Section 6.3).
6. Experiment environment (see Section 5.5.1).
7. Sample group type (see Section 5.5.3).
8. Sample size (no. of students in study) and age distribution (see Section 5.5.2).
9. Data annotations,
10. Data collection methods.
11. Used channels for affective measurement and their types (see Section 5.2).

3.4. Risk of Bias Assessment

The application of the ROBIS tool (Whiting et al., 2016) to assess the bias risk in the present scoping review on affective intelligent tutoring systems demonstrated that the methodology used is robust and well-founded. The eligibility criteria were clear and appropriate, the search strategies were comprehensive, and the processes of study selection and data collection were rigorously conducted to minimize potential errors. Since this is a scoping review, we did not conduct a quality appraisal of the included studies, which is consistent with the framework proposed by Arksey and O'Malley (2005). All these aspects led to the conclusion that there is a low risk of bias in this review, reinforcing the reliability and validity of the findings, and ensuring that the results reflect an unbiased interpretation of the available literature.

4. Related Work

Literature on affective computing for education has been reviewed before by the community. This section presents the scope of the reviews already published in the literature, their shortcomings, and how the present literature review covers the gaps. Table 2 describes the scope of the past literature reviews. As already discussed, the most recently published review papers only cover literature up to 2019 (Hasan et al., 2020), while the period from 2000 to 2010 was particularly well reviewed. Increased interest in computer-based learning and students' emotional well-being during COVID has led to higher research output on affective computing in recent years, making this review timely.

A number of bibliometric databases have been utilized by the selected review articles. Google Scholar is a service provided by Google that collates articles from various bibliographic databases with no curation or quality control. This is not beneficial for more targeted analysis, but suitable for certain bibliometric analyses. Henritius et al. (2018) selected exactly four journals, which highly limited the scope of the review.

Table 3 describes the research questions tackled in the selected review articles. Some articles reviewing specific literature have specific aims like studying the methods to measure learners' emotions in ITS (Mahmoud, 2019), studying the utility of wearable devices to capture emotions (Ba & Hu, 2023), investigating the statistical classifiers for emotions (Lek & Teo, 2023), while others like Yadegaridehkordi et al. (2019) are broader and cover a wide range of questions regarding trends in the domain. D'Mello (2013) provided a meta-analysis on the incidence of affective states during learning with technology, which is also very specific. It numerically establishes that discrete states like engagement, boredom, and confusion are the most frequently occurring learners' states.

Table 2. Scope of past literature reviews.

Article	Period	Databases	Size
Vogel-Walcutt et al. (2012)	1980–2011	Google Scholar, PsycINFO	109
C.-H. Wu et al. (2015)	1997–2013	ScienceDirect, Web of Science Core Collection	90
Henritius et al. (2018)	2002–2017	*British Journal of Educational Psychology*, *British Journal of Educational Technology*, Interactive Learning Environment	91
D'Mello (2013)	2007–2012	ERIC, PsycINFO	24
Malekzadeh et al. (2015)	2008–2014	IEEE Xplore, ACM Digital Library, ERIC, ScienceDirect, SpringerLink, Web of Science Core Collection	8
Yadegaridehkordi et al. (2019)	2010–2017	Web of Science Core Collection, ScienceDirect, IEEE Xplore, SpringerLink	94
Hasan et al. (2020)	2014–2019	ACM Digital Library, WoS, IEEE Xplore, SpringerLink, ScienceDirect	148
Khadimallah et al. (2020)	2016	IEEE Xplore, Google Scholar, Elsevier, Scopus, ScienceDirect, SpringerLink	49
Ba and Hu (2023)	2008–2022	IEEE Xplore, Elsevier Scopus, ERIC, Web of Science Core Collection, PsycINFO, ACM Digital Library, LISA, LISTA	50
Lek and Teo (2023)	2018–2022	IEEE Xplore, Google Scholar, Elsevier, Scopus, ScienceDirect, SpringerLink	48

All selected review articles except (Arguel et al., 2017; Mahmoud, 2019; Zhang et al., 2022) are systematic reviews. The scope of a review is significantly moderated by the search query used to extract a review corpus from bibliographic databases. The most frequently used search query terms (by these past reviews) are "affective computing in education", "affective computing in learning", "emotions", "emotion recognition", "intelligent tutoring system", and "emotion in learning". These terms efficiently cover the intended domain, while some studies also use more specific terms like "tutoring system" and "e-learning". However, terms like "tutoring systems", "e-learning", and "wearables" are only associated with a small subset of research performed in the entire domain of affective computing for learning education, narrowing the scope of the reviews that use these terms.

For the reasons stated before, a gap is seen in the literature as there seems to be no general systematic review of a 10+ year period/corpus post-2010 to analyze the trends and achievements of affective computing in the education domain. Hence, the current paper focuses on reviewing the period from 2010 to 2023.

Table 3. Research questions of past reviews.

Article	Research Questions
Malekzadeh et al. (2015)	Emotional regulation in ITS
C.-H. Wu et al. (2015)	What are the trends? What were the research sample groups? What are the learning domains? What are the main problems encountered?
Arguel et al. (2017)	Methodological approaches for detecting confusion
Henritius et al. (2018)	What concepts have been used in virtual learning? Who are the key theoretical contributors? How is virtual learning applied?
Yadegaridehkordi et al. (2019)	What are the trends? What are the research purposes and learning domains? What are the measurement channels and methods? What are the theories/models of emotion adopted?
Mahmoud (2019)	Methods to measure learners' emotions in ITS
Khadimallah et al. (2020)	What is the impact of emotion on learning? What are the tools for emotional regulation? What are the strategies for emotion management? Are there any emotion regulation datasets?
Ba and Hu (2023)	What wearable devices are adopted? Which emotions can wearables detect? What are the limitations of wearables?
Lek and Teo (2023)	Which emotion-classifying algorithm is used? Are classifiers real-time? What is the accuracy?

5. Systematic Review

5.1. RQ1: What Are the Learning Domains?

The selected articles study learners' affect in various learning domains. Each learning domain encompasses various subjects that activate different cognitive abilities of the brain, presenting unique affective challenges. This makes the learning domain a significant factor to consider. The total number of articles that study a specific learning domain is 48, which is approximately a quarter of the articles in the entire corpus.

The most studied learning domain is second-language (L2) learning. Second-language learning achievement can be highly dependent on the motivation, fear, and shame of students (X. Wang & Li, 2022) as technological advancements and globalization reduce the need for people to learn a foreign language. L2 learning is also highly influenced by cognitive abilities like vocal and auditory discriminating abilities, attention, recall, and memory. Most of the articles, like (Wang, 2022; Q. Wang & Jiang, 2022), study English as a foreign language (EFL).

The next most popular learning domain in the corpus consists of STEM (science, technology, engineering, and mathematics) courses as shown in Figure 2. STEM courses like math, engineering, and chemistry require higher numerical, verbal, and general reasoning abilities, while spatial cognition also plays a major role in fields like physics, biology, and engineering (Jarrell et al., 2022). In Henry et al. (2021), the fear of failure in STEM learning is quantified through statistical means while anxiety is studied by a variety of articles like (C.-M. Chen & Hong, 2010; Karimi & Fallah, 2021; M. Liu & Hong, 2021) in the context of L2 learning.

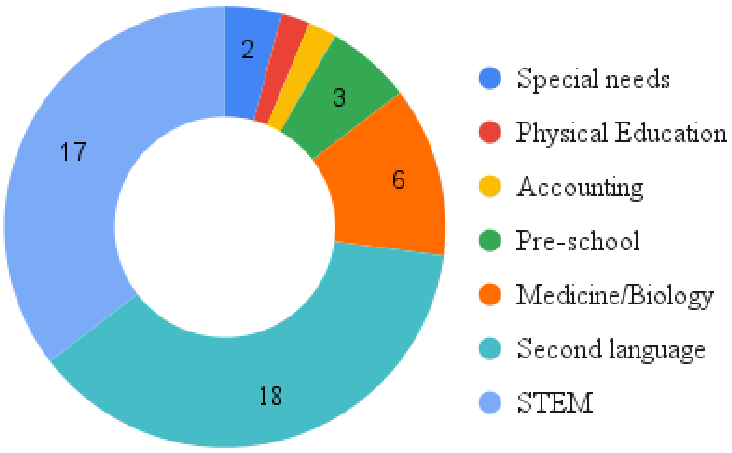

Figure 2. Learning domain distribution.

Many articles, like (Behrens et al., 2019; Naeem et al., 2014), studied the affect and cognition of nursing, doctor, and physician students in the medical domain. Particularly associated with these fields are students' perceptions of disgust and the emotional effects of being close to dying beings. Two articles from the corpus investigated affect in the learning domains of special needs education (Andrunyk & Yaloveha, 2020; Zhang et al., 2022); one article focused on physical education (Alcaraz-Muñoz et al., 2020), and another focused on accounting (H.-C. Lin et al., 2014). A limited number of articles studied preschool education, like (Alamos & Williford, 2020).

5.2. RQ2: What Affective Measurements Are Used?

An extensive variety of affective measurements are reported in the selected articles. The authors follow the categorization of affective measurements described by (Yadegaridehkordi et al., 2019) and shown in Figure 3.

1. Textual: These affective measurements involve manual input from the learner or the researcher. Emotion questionnaires are some of the most popular channels for affective measurement. A total of 78 different questionnaires are reported in the selected corpus. The most frequently used questionnaire is the Achievement Emotions Questionnaire, described in detail in RQ15 as a seminal work in the domain. While studying emotion questionnaires could be the subject of another review altogether, the frequently used questionnaires are mentioned briefly. The positive and negative affect schedule (PANAS) is used to measure the level of positive and negative emotions of learners (Münchow & Bannert, 2019). The intrinsic motivation inventory (IMI) (Ryan, 1982) is used to measure the level of motivation of learners. The student–teacher relationship scale (STRS) uses Likert-type questions (Berhenke et al., 2011) to measure the closeness and conflict factors in student-teacher relations. Similarly, the motivated strategies for learning questionnaire (MSLQ) is used to measure the learning strategies used by college students (Cho & Heron, 2015). The academic emotions regulation questionnaire (AERQ) (Alazemi et al., 2023) is used to measure the ability of students to regulate their academic emotions. The questionnaire is administered at regular intervals during lessons. Other textual channels include interviews (group and personal), expert observations (usually performed in clinical settings), and manual text entries like emotion diaries and prompts.
2. Visual: This affective channel comprises visual data like facial expressions, gestures, and eye tracking. This channel has been very popular since the implementation of data-collection methodology is easy and non-invasive while the data are rich in

depth of affective measurements since humans express a wide range of emotions through visual cues. Facial expressions are the most popular kind of affective data after questionnaires. The facial action coding system (FACS) (Ekman & Friesen, 1978) is the founding standard facial emotion taxonomy used in numerous articles (G. Li & Wang, 2018; Mano et al., 2019; Stepanov et al., 2019; Tonguç & Ozaydın Ozkara, 2020; Vassiou et al., 2015). Body gestures are used once (Kanimozhi & Raj, 2017) and, eye-movement tracking is reported four times (Fwa, 2018; H.-C. K. Lin et al., 2022; Singh et al., 2022; W. Yu et al., 2022).

3. Vocal: Only one vocal channel for affective measurement is reported—vocal intonation. The pitch, intonation, intensity, and loudness of speech can reveal important affective information. Vocal intonation is used by Tu et al. (2012) to automatically assess boredom in learners. Similarly, vocal intonation is used by H. Chen et al. (2017) to analyze the affect of learners in mobile learning environments. Jie et al. (2020) propose a unique approach of recording the teachers' vocal intonations to deduce their emotions.
4. Physiological: These channels for affective measurement tend to be the most invasive, and are usually reported from limited laboratory environments, as it is difficult to scale their usage for multiple learners. The most frequently reported physiological signal used is the heart rate/pulse. An electroencephalogram (EEG) is reported four times in the corpus. Brain activity revealed through the EEG is probably the most accurate channel for affect determination as described by Babiker et al. (2017) and Shen et al. (2014). The galvanic skin response (GSR) is an interesting channel endemic to emotion research. Changes in sweat gland activity are measured to indicate the intensity of emotions (J. Li et al., 2020). Blood pressure has been used once (Ray & Chakrabarti, 2016) to enhance technology-enabled learning. Popescu et al. (2018) describe a scalable smart classroom technology that uses students' seats' pressure to deduce affect.

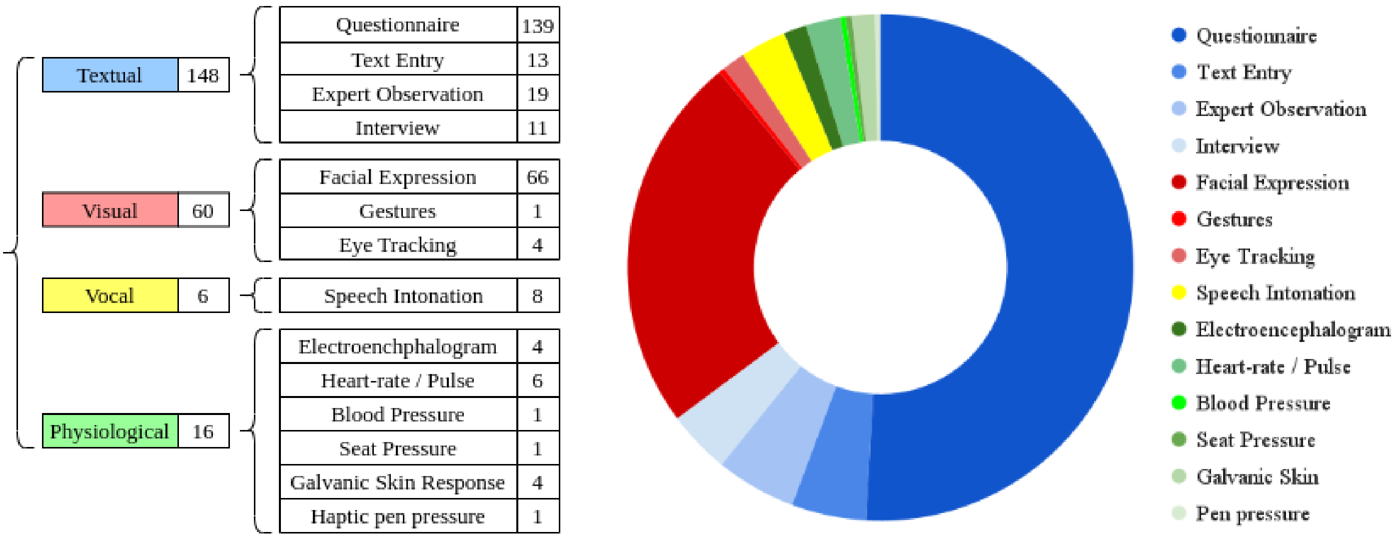

Figure 3. Distribution of channels for affective measurement utilized.

Worth mentioning is a unique approach reported by Schrader and Kalyuga (2020) to evaluate student emotions using haptic feedback from a special stylus' used in conjunction with touchscreen devices.

5.3. RQ3: What Datasets Are Available?

Most articles in the corpus utilize statistical methods like machine learning, structural equation modeling, and statistical indicators to prove their hypotheses or test the accuracy of the proposed affect recognition systems. In such a scenario, datasets play an essential role in increasing the accuracy of systems, revealing characteristic trends in the sampled population, and reproducing results.

Within the selected corpus, four articles specifically propose datasets for affective computing in education. All four datasets contain data related to the facial expression affective channel.

1. Hu et al. (2020) proposed RFAU (real classroom facial action unit), a database of 256,220 photographs of 1796 Chinese juvenile students and teachers in real classroom scenarios. The images are annotated by qualified annotators with 12 action units and their intensities. The database is intended to train algorithms to identify the emotions of students in classrooms through their facial actions.
2. BNU-LSVED 2.0 (Beijing Normal University Large-scale Spontaneous Visual Expression Database version 2.0) was proposed (Wei et al., 2017). It is a database of 2117 video recordings of 81 college students. The videos are self-annotated by the students using categorical emotion labels, affective behavior/gesture labels, and a pleasure–arousal–dominance (PAD) scale label. The database is advantageous because students are seated randomly, which adds significant variation to each student's videos.
3. The student engagement dataset contains 18,721 images of 19 college students solving mathematical problems. It was proposed by (Delgado et al., 2021). The dataset annotations were crowd-sourced through the Amazon Mechanical Turk outsourcing platform. The head-pose and gaze of students were annotated with 3 labels: "looking at the screen", "looking at the paper", and "wandering".
4. OLFED-HO (online learners' facial expression with hand occlusion) is a dataset of 92,497 facial images of 30 Chinese undergraduate students. The dataset is unique in that it has been annotated with labels describing occlusion, i.e., hiding or covering, of the face by the learners' hands. This is particularly important because learners often display emotions through hand gestures, and also occlusions decrease the accuracy of a statistical classifier by hiding facial features. The dataset is further annotated with the seven academic emotions (Lyu et al., 2022).

Data from none of the four datasets is publicly available. No response has been received from the corresponding authors when inquiring about the datasets.

The selected articles have collectively reported 16 other emotion measurement datasets. The data comes from different channels, including facial expressions, speech, text, questionnaires, etc. The data sets reported in multiple articles are briefly described. The Facial Expression Recognition 2013 (FER2013) database is the most popular dataset (https://datasets.activeloop.ai/docs/ml/datasets/fer2013-dataset/ (accessed on 15 November 2023)). It contains 30,000 RGB images of the faces of people annotated with seven emotions based on the FACS. The JAFFE (Japanese Female Face Expression) dataset (Lyons et al., 2020) contains 213 images of ten Japanese female subjects with the same FACS annotations. The CK+ (extended Cohn–Kanade) database (Lucey et al., 2010) contains 593 video recordings of 123 subjects. In each recording, the subjects transition from a neutral affect to one of seven emotions according to FACS, with varying intensities. The Radboud Faces database (Langner et al., 2010) contains 5880 images, at varying angles of the face and eyes, annotated according to FACS. CASIA (Chinese emotional speech) corpus is one of the three emotional speech datasets reported by the reviewed articles. It contains recordings of 12,000 sentences in Chinese spoken by four trained speakers in six emotions. The KDEF (Karolinska directed emotional faces) database (Lundqvist et al., 1998) is one of the oldest facial emotions datasets. It contains 4900 images of the emotive faces of 70 amateur actors according to seven emotions. Although most databases focus on facial visual expressions, some other notable datasets include the DEAP (database for physiological emotional analysis) dataset (Koelstra et al., 2012) containing 32-channel EEG and facial video recordings of 32 participants watching music videos in a laboratory environment. The data is self-annotated by the participants through the Valence, Arousal, and Dominance labels. FABO (bimodal face and body gesture) dataset contains 1900 videos of the face and body gestures of 23 participants (Gunes & Piccardi, 2006). Data were self-annotated using FACS emotions with the emotions 'anxiety' and 'boredom'. Increasing in popularity is the DAiSEE data set that contains 9068 video snippets annotated with affective states of boredom, confusion, engagement, and frustration (Gupta et al., 2016).

Most of the reviewed articles do not provide their affective data publicly, especially data recorded from questionnaires.

5.4. RQ4: What Types of Affective Systems Are Proposed?

Affective systems proposed in the reviewed articles are primarily of two kinds: emotional recognition and emotional regulation.

A total of 17 articles propose emotion recognition systems. These systems take one or more channels for affective measurement as input and output fixed values of the subject's detected affect. These systems are not intended to make a difference in the subject's affect but instead to ultimately better augment emotional regulation systems. Many of these systems are designed to indicate the learners' emotions to teachers, which can help teachers adapt their teaching style to better suit students' needs. For example, Dewi et al. (2011) used local binary pattern-based ML to achieve the same, while Ez-zaouia et al. (2020) used the Microsoft Emotion Detection API to achieve the same for the online environment. Arguedas et al. (2016a), Soltani et al. (2019) and Tsai and Lin (2021) propose feedback systems that directly notify the students about their affective states, expecting them to self-regulate their emotions more effectively. On the other hand, Arguedas et al. (2016b) proposed an algorithm to label students' emotions in computer-based learning systems.

A total of 40 articles propose emotional regulation systems. These are systems capable of directly influencing the affect of learners. Emotional regulation systems are interventions that take many forms, for example, emotion-aware reinforcement systems aim to shift the learners' affective state from negative emotions to pro-learning emotions through various means, like changing the tone/expression/emotion of the tutoring agent (Gu et al., 2010; Moridis & Economides, 2012; Rathi & Deshpande, 2019), suggesting activities to increase the positive emotions of the students (Landowska, 2013), or changing the course materials to suit the emotional state of students (Moga & Antonya, 2012; Mohanan et al., 2017). The most common approach involves changing the emotions of the automated affective agent. This approach closely mirrors real-life classroom scenarios, where the human teaching agent alters their emotions in response to the students' affective and cognitive states. Notably, Cabada et al. (2015) proposed an ITS based on the Java platform, H.-C. Lin et al. (2012) used both facial recognition and text-based affect recognition to identify and concurrently regulate learners' emotions. Game-based learning, as described by Barz et al. (2023), is based on the hypothesis that games can contribute to inducing positive academic emotions in learners and make course material more interesting, which reduces boredom.

5.5. RQ5: What Examinations Are Conducted to Validate Interventions?

Various kinds of examinations are reported in the reviewed articles. The kind of experiment depends on the goals of the article. Alepis and Virvou (2011) and Ma et al. (2018) proposed emotional regulation or emotion recognition systems based on questionnaires and automatically collected data to perform quantitative examinations. Articles like these use statistical methods like regression and correlation analyses to identify the accuracy of the proposed system. More so than not, studies, such as by (Ozek, 2018), performed a qualitative analysis rather than a quantitative one, in the form of collecting affective feedback from users of their platforms. These articles base their hypotheses on empirical methods. Articles proposing new theoretical models of emotion, such as by (Pelch, 2018), or that study the relationship between different affective and cognitive concepts, such as by (Brooks & Young, 2015), solely rely on questionnaires and student grade-tables to perform quantitative statistical analyses. They perform confirmatory-factor analyses and structural equation modeling to identify the correctness of their models and the strength of relationships between the concepts that they analyze.

A qualitative study is important when considering end-usability, the design of the user interface, and general/professional judgment of the value of the proposed system, as performed by Y. Yang et al. (2022). A quantitative study, on the other hand, is useful when the project focuses on modeling the behavioral patterns of learners (these patterns could be in the form of patterns of externalizing behavior (AlZu'bi et al., 2022), or the internal effects of cognitive and affective parameters, as described by X. Wu et al. (2023)). Both approaches provide complementary insights, depending on the goals and scope of the research, and the choice between them should be guided by the specific research questions and objectives.

In the present research corpus, approximately half of the studies that had a research purpose in the 'designing an emotion recognition/expression system/method' category (see Section 6.4), performed a qualitative analysis using questionnaires, while the other half performed/reported quantitative analysis based on precise numerical metrics.

In the following sub-research questions, the various features of the examinations, experiments, and validation techniques encountered in our corpus, are discussed in greater detail.

5.5.1. RQ5.1 What Are the Environments in These Examinations?

The environment of the learner plays a vital role in their affective state and emotional regulation. The authors identified three distinct categories of learning environments from the selected articles based on the amount of control over environmental factors.

1. 'Laboratory' is an environment with the highest amount of control. The learner is often completely focused on the learning task since there are no (unplanned) distractions or interferences in a controlled laboratory environment. This environment is not realistically reproducible for a large number of learners. While it is beneficial for theoretical research purposes, non-academic emotions induced by distractions are minimal, allowing for a clean measurement of academic emotions (Ali & Hamdan, 2017; C.-M. Chen & Sun, 2012; Lehman & Zapata-Rivera, 2018; Ruiz et al., 2020).
2. 'Classroom' is the typical school environment, where groups of learners are taught by single or multiple teachers. In this case, the environment is semi-controlled, and more factors are at play considering learners' interpersonal interactions and relations. Classroom dynamics are often analyzed to identify their effect on the learning process (Beetz, 2013; Bellocchi & Ritchie, 2015; C.-C. Chung et al., 2019; Feidakis et al., 2013).
3. 'Online' represents the learning environment with the least amount of control. With the widespread curfews witnessed in 2019–2021, online learning became the de-facto environment. In such a scenario, the learner learns through an Internet-enabled device

from a location of their choice. While the learners benefit from added fidelity, they are susceptible to various environmental interference. In such a de-personalized scenario, attention and emotional regulation are often the subject of articles studying this learning environment (Aslan et al., 2018; S. Chung et al., 2015; Daouas & Lejmi, 2018; Hsu et al., 2014).

Figure 4 presents the distribution of learning environments within the selected articles. Classrooms are the most researched learning environments, followed by online and laboratory environments. This could be because the web-based online environment is relevantly new and facilitated by technological advancements, while the laboratory environment is only helpful for theoretical results and does not represent real-life scenarios.

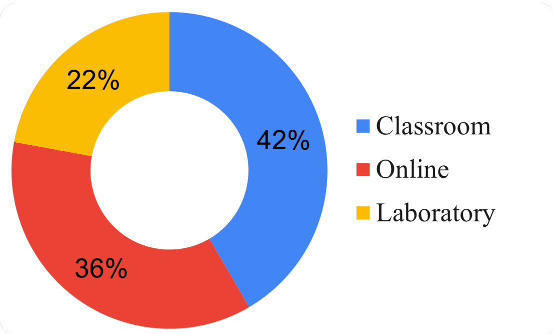

Figure 4. Learning environment distribution.

5.5.2. RQ5.2 What Are the Sample Sizes and Ages in These Examinations?

Sample sizes vary widely in between studies. Daher and Swidan (2021) considered only 8 participants in the experiment, while Yılmaz et al. (2022) collected data from 3655 participants to validate their hypothesis. Figure 5 presents a histogram of the sample sizes of the reviewed articles, where the size is less than 500. 13 articles have sample sizes greater than 500 with a maximum sample size of 19,044 reported by Ainley and Ainley (2011). The sample size is directly proportional to the confidence in a study's results. However, the sample size is limited for feasibility. Often it is more economical to administer self-reported questionnaires to learners than to collect digital data like video or EEG signals effectively.

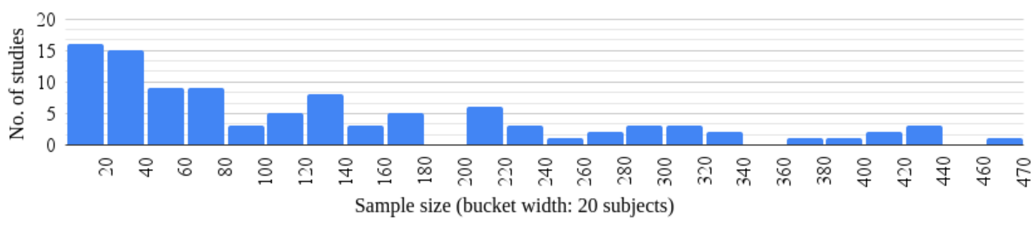

Figure 5. Sample size histogram (for size <500).

Figure 6 describes the reported sample age groups in the reviewed articles. The age and range vary significantly according to the learning domain, environment, aim, and scope of the article. As can be seen from the figure, the age group from 15 to 25 years is the most studied. This is expected since (as is described in the answer for RQ5.3), the most popular sample groups are undergraduate and high-school students.

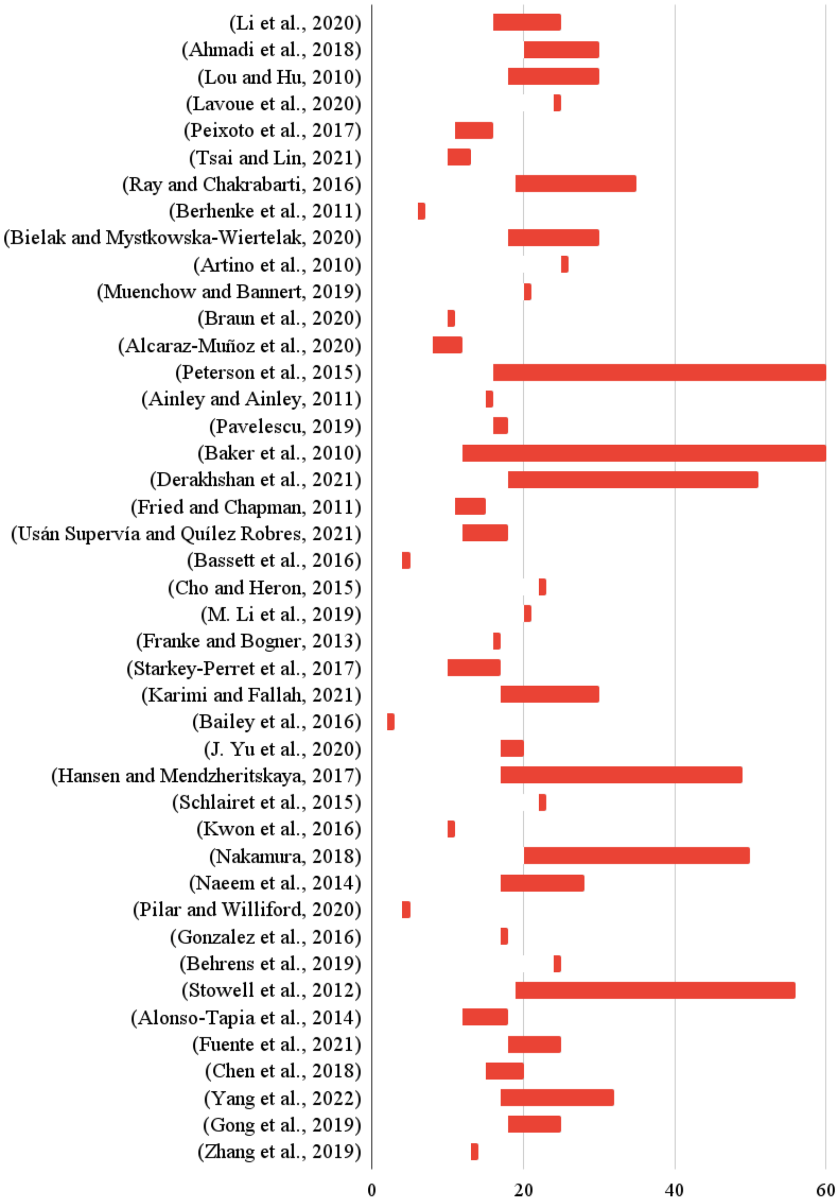

Figure 6. Sample age range of learners in reviewed articles. Note: Given the extensive number of references involved in data extraction, we have chosen not to list them alongside the figure captions. This convention is applied across the subsequent figures. All relevant sources are properly cited within the main text.

5.5.3. RQ5.3 What Are the Sample Groups in These Examinations?

The reported examinations cover a wide range of learner sample groups, as shown in Figure 7. The most popular sample group is undergraduate students, accounting for almost 60% of all reported samples. This is expected since undergraduate students are usually the most numerous and immediately available population in educational research institutions. High school students are the next most popular researched group followed by graduate students. There are some studies based on preschool students and online learners. Notably, Andrunyk and Yaloveha (2020) studied the requirements of special-needs students, Berhenke et al. (2011) studied pre-school head-start students (a US-federally funded program designed to promote school readiness for children aged three to five from low-income families), and Zhao and Wang (2023) studied the interplay between grit and emotion in ethnic minority students in China. The only article dealing with a non-academic sample group is by Baldassarri et al. (2013), which experiments with corporate employees to validate their proposed system.

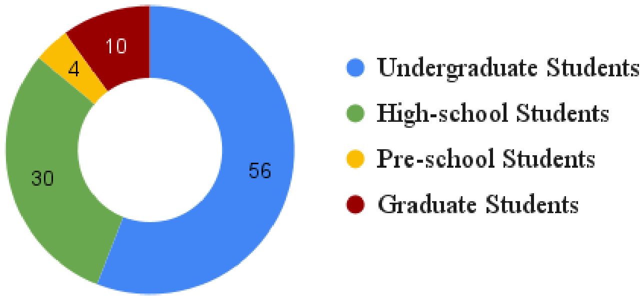

Figure 7. Sample group distribution of reviewed articles.

5.6. RQ6: How Well Do the Interventions Perform?

Figure 8 graphically presents the reported accuracies of interventions in the articles of the review corpus. The best accuracy (99.1%) among all reviewed articles is reported by F. Wang (2022) to detect the intensities of 9 different emotions in the ADFES-BIV database described in RQ3. They achieve this using deep-learning methods. Afzal and Robinson (2011) report an accuracy of 95% in recognizing six different emotions through the proprietary Neven Vision face-tracker software to identify facial action units. Then, a bank of hidden Markov model-based modules is used to learn the temporal variations in data, for classifying emotions.

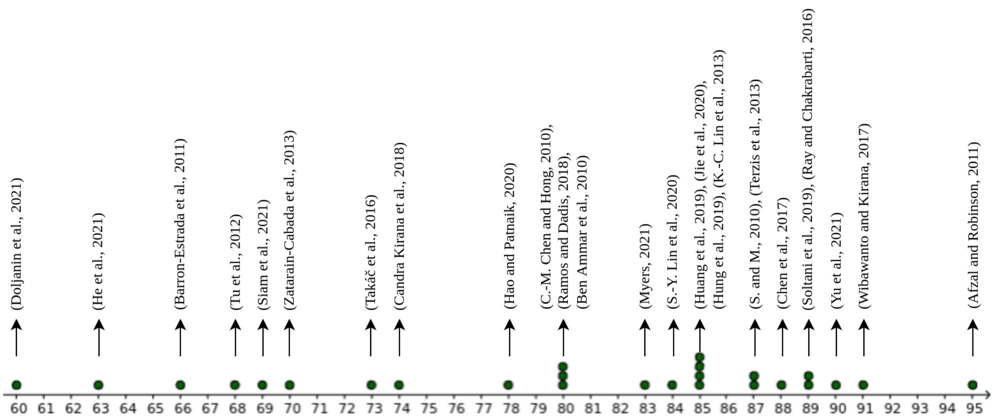

Figure 8. Accuracy distribution of interventions in reviewed articles.

Other notable works include (Wibawanto & Kirana, 2017), who achieved 91% accuracy by using a Fisher Face-based emotion recognition method and a backpropagation algorithm to classify student emotions. H. Yu et al. (2021) reported an accuracy of 90% by utilizing emotion recognition integrated with the improved AprioriTID algorithm to evaluate online teaching quality. Soltani et al. (2019) and Ray and Chakrabarti (2016) achieved 89% accuracy, both incorporating emotion recognition into learning systems with a combination of facial expressions and biophysical signals. Finally, H. Chen et al. (2017) reported an 88% accuracy in emotion recognition, focusing on constructing an affective education system for mobile learning environments.

The mean accuracy of the quantitative studies of (affective computing-based) academic interventions in our corpus is 80.74% (with a standard deviation of 10.2%) while the median accuracy is 82.75%. The lowest accuracy was reported by Doljanin et al. (2021), even though they use not just facial expressions but head and body posture data as well. The average accuracy is believed to be very good, considering the industry in affective computing for education is still nascent. This is a good indicator of the scientific understanding of the features that affect academic emotions in learners. Learning specific data is not very easy as there is a limited number of datasets in this field, as shown in RQ3. However, as more data are generated and shared, new inferences that increase the accuracy of emotion detection models can be made.

5.7. RQ7: Which Theoretical Models of Emotions Are Reported?

The reviewed articles have used various theoretical models of emotions as the basis of their research. These models are theories that clearly define the static and dynamic relationships between different affective or cognitive concepts. A total of 30 theoretical models of emotion have been reported in the reviewed articles. The multiply-used and other noteworthy theoretical models of emotion will also be discussed.

The control-value theory of achievement emotions (CVTAE) model (Pekrun, 2000) is probably the most popular model within the domain, having been reported eight times within our corpus. The theory states that the level of control and the value of the outcomes stemming from achievement-related activities are the most critical factors in determining achievement-related emotions. It also states that a student's emotions play an essential role in determining their cognitive abilities and academic performance. This is but a very small part of the overall model, which covers a lot of scenarios and factors influencing and being influenced by the learners' emotions.

The cognitive affect theory of learning with media (CATLM) model (Moreno, 2006) is an important theoretical model of emotion today, as it bridges the gap between cognition and affect. It assumes that learners' cognitive engagement is mediated by their motivation, cognitive and affective processes depend on meta-cognition, and that prior knowledge affects the efficiency of learning. It draws from the original cognitive theory of multimedia learning (CTML) model. CTML lays the foundation for understanding how students process information through various channels offered by multimedia techniques.

The self-determination theory (SDT) (Deci & Ryan, 1985) is an important theory that describes the subject's intrinsic motivation. Applied to the current context, this theory is used to explain the intrinsic academic motivations of students to read and learn. For example, it is used by Y. Chen et al. (2018) to understand the role of goal orientation and help-seeking vis-à-vis shyness and adjustment, which relate to SDT's concepts of autonomy, relatedness, and competence.

The social-cognitive theory (SCT) (Bandura, 1986) describes the relationship between the subject's cognition, behavior, affect, and social environment. This theoretical model of emotion explains the reciprocal relationship between the subject, environment, and behavior. SCT incorporates the concept of reinforcements, which relate to the internal responses to the subject's behavior and affect the likelihood of continuing the behavior. It is unique in including the concept of self-efficacy, which is a person's confidence in their abilities. Such a theoretical model of emotion is advantageous when studying the role of interpersonal and social relationships in a classroom environment on students' cognition, as demonstrated by Bown and White (2010).

The facial action coding system (FACS) (Ekman & Friesen, 1978), while not being a concrete theoretical model of emotion, is the most reported emotion taxonomy in the reviewed articles. It categorizes the physical expression of emotions through the face by defining atomic action units (contractions or relaxations of specific facial muscles). The action units are related to seven basic emotions: happiness, sorrow, surprise, fear, anger, disgust, and contempt. It is the most widely used facial coding taxonomy. While the taxonomy is very useful for emotions, it is also used in various other domains. Most facial expression recognition systems follow the FACS guidelines.

Recently, there has been a surge in the number of articles basing their approach on the Well-being Theory of Positive Psychology proposed by Seligman (2018). They believe that well-being is made of five elements: positive emotion, engagement, relationships, meaning, and accomplishment (PERMA). This is based on a more holistic approach toward learning that focuses on learners' beliefs about their goals and achievements. The approaches advocate for a pedagogical change that better relates learning goals to learners' emotions.

5.8. RQ8: Which Emotions Are Reported?

A total of 48 different emotions have been reported in the reviewed articles. Figure 9 presents the distribution of the emotions reported at least 10 times. The most reported emotions comprise the seven basic emotions associated with the FACS, namely, joy, anger, sorrow, fear, surprise, neutral, and disgust. Boredom and anxiety, which are important emotions in the educational context, also frequently occur. Often, various synonyms are used to represent the same emotion. For example, joy, enjoyment, delight, and pleasure are synonyms of the emotion called happiness. Emotions like engagement, confusion, frustration, pride, shame, and hopelessness are also frequently reported. Disgust is often studied in the medical domain to understand the effect of dissection on the academic emotions and achievement of medical students (Kaiser et al., 2023).

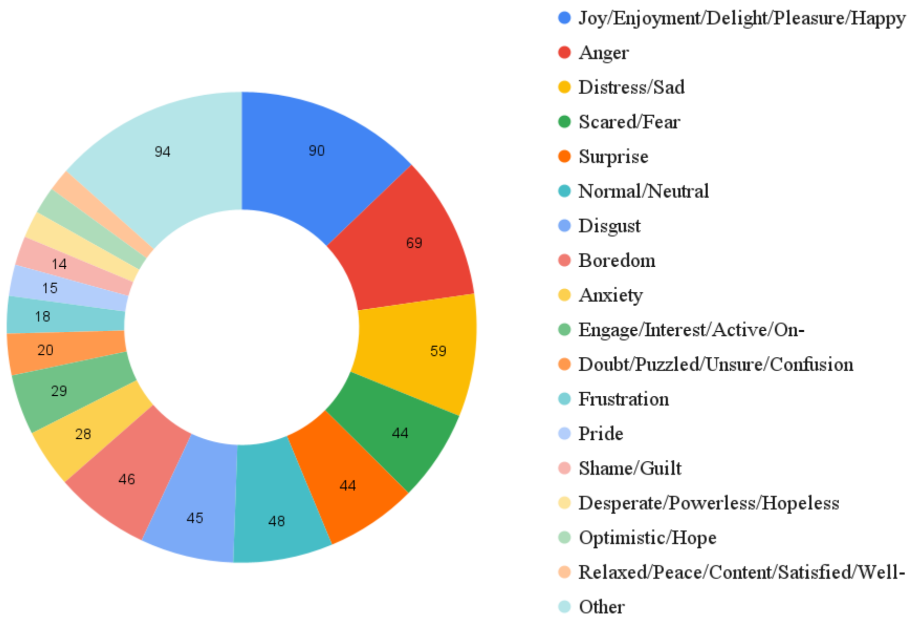

Figure 9. Emotions and frequency reported in reviewed articles.

Often, valence and arousal are studied instead of discrete emotions (S. Chung et al., 2015). Unique combinations of valence and arousal may be interpreted as distinct emotions. Also, some articles study the broad categories of 'positive' and 'negative' emotions instead of discrete emotions (Satyam et al., 2022). Positive emotions are associated with better cognitive performance, which is desirable, while negative emotions hinder performance and quality of life. The semantics of the emotion terms can vary in between studies. For example, while there may be no recognizable difference between enthusiasm and flow, the terms are treated as distinct affective states by J. Li et al. (2020), to measure varying levels of happiness and excitement.

6. Data Analysis

6.1. RQ9: What Are the Temporal Trends of the Domain?

Figure 10 illustrates the number of articles published every year from 2010 to 2023 within the reviewed pool of 175 articles. The average number of articles published from 2010 to 2016 is 8.4. The average number of articles published in the last 6 complete years, i.e., from 2017 to 2022, is 18, more than double the period from 2010 to 2016. Figure 10 also illustrates the exponential trend line, which is an increasing function.

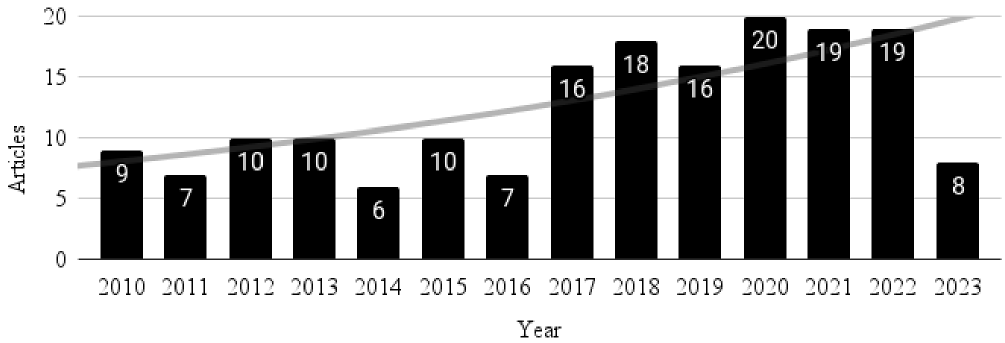

Figure 10. Temporal trends of reviewed articles.

The COVID-19 pandemic shifted the primary mode of instruction from physical in-person lectures to online video and assessment-based lectures. An online mode of education is not able to satisfy the affective needs of all stakeholders, and neither can physical in-person tuition. This pushed research to fill the gaps and overcome the affective burden faced by students and teachers from this shift. This is inferred from the increasing publications in the domain within the last 3 years.

Given the increasing number of publications in the field, particularly from 2017 to 2022, and the ongoing advancements in the digitization of education, the authors predict that the trend of increased interest in affective computing solutions for education will continue. Research in this domain has not matured enough yet, and the current major change in education—the shift toward digital classrooms—provides a strong argument for the need for affective computing. As shown through other research questions in the systematic review, there is no universally accepted emotional model or affective learning tool as of yet. However, preliminary data for many tools appear promising. Based on these observations, the authors predict even greater interest in affective computing in education in the period from 2024 to 2030. The authors believe that affective computing in education is still in Phase 1 (Technology trigger) of the Gartner hype cycle, marked by initial proofs-of-concept, but lacking proven or demonstrated commercial viability.

6.2. RQ10: What Are the Venues for Publication of Research in This Domain?

As described in Figure 11, out of the reviewed articles, 50 articles were published in conferences, while 125 articles were published in journals. Usually, journals are associated with more stringent reviews consisting of multiple review and rebuttal rounds. Hence, this is an indicator of the high quality of articles selected for the review.

Figure 12 shows the accessibility of the reviewed articles: 50 of the selected articles have open access, while 125 articles require some sort of subscription to the corresponding publishers' bibliometric services for access. Open access is a recent movement, propelled by the advent of the Internet, which advocates for scientific works to be available free of cost, to increase accessibility of science, to increase collaborations, and to reduce costs. Often, open access-enabled publications have higher citation metrics due to higher visibility (it is important to clarify that the inclusion criteria for this systematic review were not influenced by the accessibility or open access status of the publications. All relevant articles, regardless of whether they were open access or behind a paywall, were considered for inclusion based on their relevance and quality).

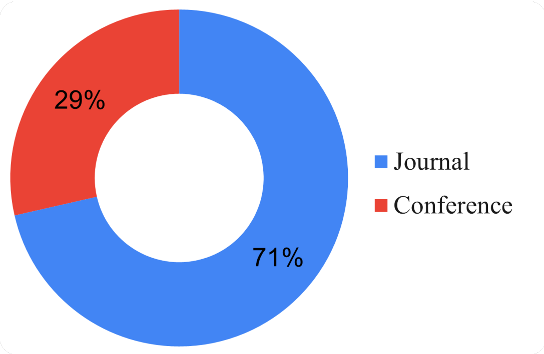

Figure 11. Publication venue distribution.

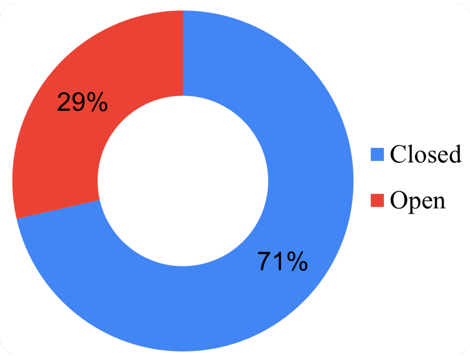

Figure 12. Availability of reviewed articles.

The selected articles are published in 79 different journals and presented in 47 different conference series. Table 4 describes the 15 most published journals from the reviewed articles and their impact factors. *Frontiers in Psychology* seems to be the most relevant journal, with 9 articles.

Table 4. Top 15 most-published journals in this review, and their 2023 2-year impact factors (IFs).

Journal	No. Articles	IF
Frontiers in Psychology	9	3.8
Computers and Education	4	12
System	4	6
Educational Technology and Society	3	4
Journal of Educational Computing Research	3	4.8
Interactive Learning Environments	3	5.4
International Journal of Human–Computer Studies	3	5.4
IEEE Transactions on Affective Computing	3	11.2
Sustainability	3	3.9
Computers in Human Behavior	2	9.9
Contemporary Educational Psychology	2	10.3
Educational Technology Research and Development	2	5
Emotion	2	4.2
IEEE Transactions on Learning Technologies	2	3.7
International Journal of Educational Technology in Higher Education	2	8.6

6.3. RQ11: What Is the Quality of Published Work?

To assess the quality of the reviewed articles, the authors analyzed each of the reviewed articles according to eight questions that cover rigorousness and credibility. The authors do not include questions on the relevance of the selected studies since they are highly speculative. The quality questions (QQs) are adapted from (Mittal & Prince, 2022). Each QQ receives a dichotomous answer (yes/no), and each 'yes' answer corresponds to one point in the analysis. In the end, a general quality score ranging from 0 to 8 is generated for each study. The QQs are as follows:

QQ1.Is there a clear statement of the aims of the research?
QQ2.Is there an adequate description of the context of the research?
QQ3.Is the research design appropriate to address the aims of the research?
QQ4.Is the complete implemented design provided?
QQ5.Is there a control group to compare the proposed treatments?
QQ6.Are the analysis metrics appropriate for addressing the research issue?
QQ7.Is the analysis sufficiently rigorous?
QQ8.Is there a clear statement of findings?

The distribution of quality scores is described graphically in Figure 13. The average quality score for the 144 reviewed articles is 5.38, indicating moderate to good research quality in the domain. The quality score distribution follows a natural distribution. The central limit theorem strongly suggests the quality scoring was random and fair. When articles lack specific information regarding their interventions or do not provide methods to access their data, it severely impacts the reproducibility and verifiability of the contributions. This quality of the work corresponds to QQ4. Most of the selected articles scored poorly in this metric.

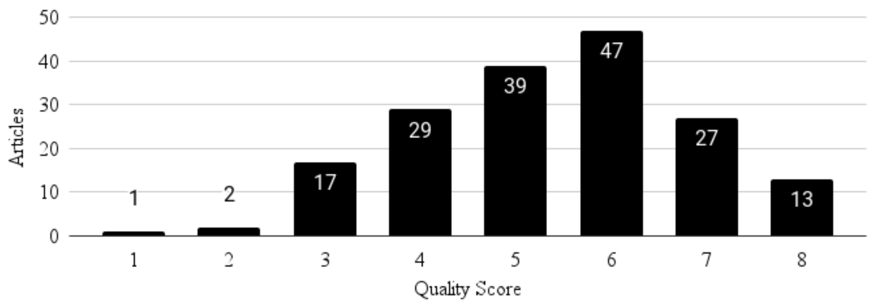

Figure 13. Quality score distribution of reviewed articles.

6.4. RQ12: What Is the Research Purpose of the Published Work?

After the three independent review rounds, all the selected articles are studied to group them according to their overall research purpose (RP). The research purpose categories are derived from those reported by Yadegaridehkordi et al. (2019). The distribution of articles according to their research purpose is described graphically in Figure 14. The categories are as follows:

- Investigate the effect of emotion on performance: These papers investigate the uni-directional effect of emotion on students' academic performance. Often, such studies utilize a questionnaire/system to track student/user emotions coupled with an assessment/system to determine their performance.
- Investigate the relationship between emotion, motivation, and learning style: These papers investigate the multi-dimensional relationship between emotions, motivations, and learning styles. Such studies often couple multiple emotion and cognitive models, proving the correctness of this coupling through data.

- Investigate the effect of lecturer behavior/material in student/user emotions/ motivations/outcomes: Such studies investigate the role of lecturer emotions, teaching style, or teaching materials in students' educational outcomes.
- Designing emotion recognition/expression system: The largest class of studies to design an emotion recognition system to capture and appropriately process students' emotions. Often, these studies aim to make the computer-based tutor more affective and reactive to students' dynamic emotions.
- Review: The class of papers reviewing the domain of affective computing in education.
- Dataset: These papers describe generated data, which is useful for studies on affective computing in education. They often contain classroom recordings, EEGs, and other physiological signals, with affective and cognitive annotations.

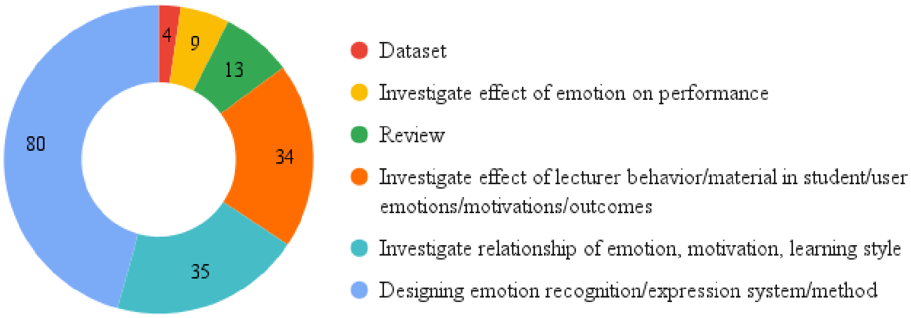

Figure 14. Research purpose distribution of reviewed articles.

Almost half (46%) of the selected articles are concerned with designing emotion recognition or expression systems. Less than a quarter (19%) of the articles study the relationship between emotion, motivation, and learning style, and a similar number of articles investigate the effect of lecturer emotions on student performance. About 6% of the articles directly study the effect of student emotions on their performance. A small number of articles are reviews or propose novel datasets. In recent years, there has been an increasing focus on investigating the effects of lecturer behavior on student emotions and their interplay with outcomes.

7. Bibliometric Analysis

Bibliometric data are collected through the dimensions AI bibliometric service (Herzog et al., 2020). Dimensions AI is one of the world's largest bibliometric databases with 129 million publications, along with grants, datasets, clinical trial reports, and policy documents, all linked by 1.6 billion citations. From the 175 reviewed articles, data for seven articles were unavailable on Dimensions AI.

The bibliometric data extracted from Dimensions AI is studied using VOSviewer (van Eck & Waltman, 2010) software. VOSviewer can generate and visualize networks of researchers, journals, publications, organizations, or countries based on citation, co-citation, bibliometric coupling, and co-authorship relations. VOSviewer can also compute the existence of various clusters, which helps infer trends in the domain. The types of networks generated by VOSviewer and inferences from them will be discussed.

7.1. RQ13: How Are the Publications Bibliometrically Connected?

Figure 15 describes the citation network of the reviewed articles. The nodes of the graph represent the reviewed articles. The edges represent links between two articles, one citing the other. The size of a node is proportional to the number of citations of the corresponding article. The link strength, i.e., the thickness of an edge, is always one since an article can cite another only once.

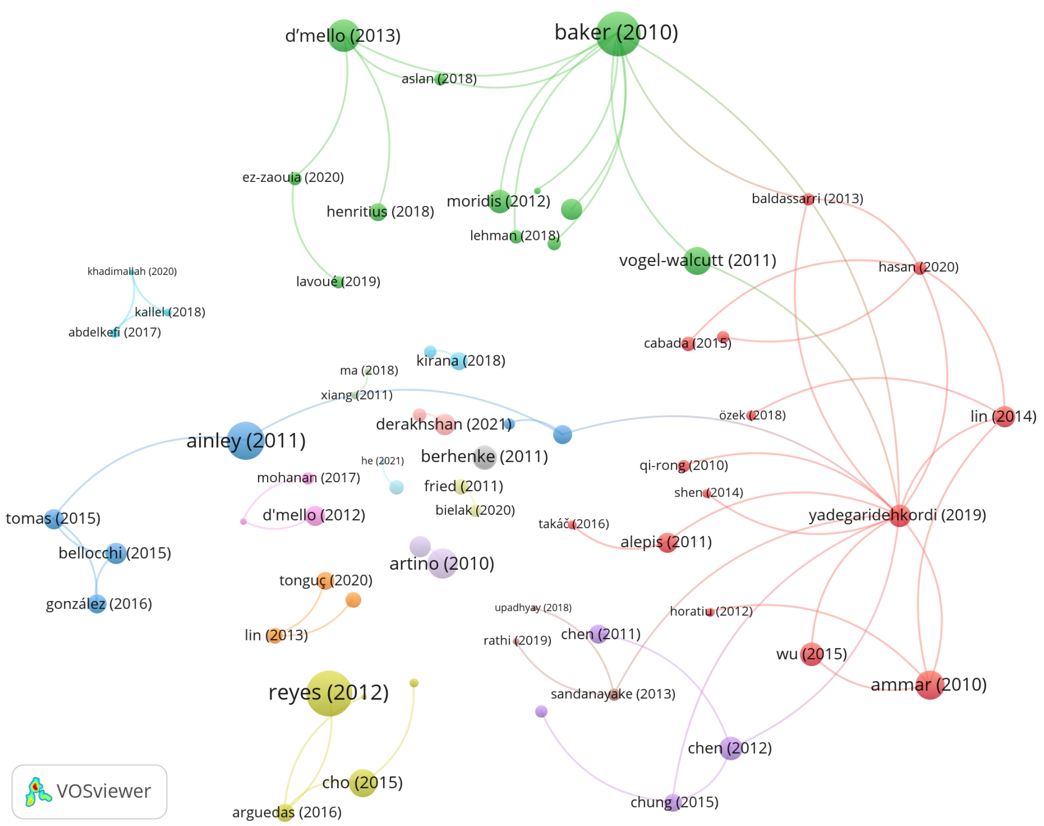

Figure 15. Citation links of reviewed articles. The normalization method used is the association strength normalization method. The resolution is 3.00, and the minimum cluster size is 2.

Table 5 describes the seven most-cited articles reviewed. The most cited article within the selected corpus is by Reyes et al. (2012) (corresponding to the largest yellow node toward the bottom of Figure 15). The article with the highest link strength, i.e., the number of connections, is by Yadegaridehkordi et al. (2019), with links to 14 articles within the corpus. In Figure 15, it is represented by the red node on the right side.

Table 5. Most cited reviewed articles.

Article	Citations
Reyes et al. (2012)	508
Baker et al. (2010)	454
Ainley and Ainley (2011)	262
D'Mello (2013)	154
Artino et al. (2010)	129
Ben Ammar et al. (2010)	106
Vogel-Walcutt et al. (2012)	104

The citation network helps visualize the dependency of the reviewed articles on each other, as well as incremental contributions to the domain, which helps to identify growing interests and trends. An article with a significant fundamental contribution will have higher citations from works building on that contribution. For example, from Figure 15, it can be observed that many articles build on the contribution of (Baker et al., 2010). Similarly, Ainley and Ainley (2011), Ben Ammar et al. (2010), D'Mello (2013) and Reyes et al. (2012) are high-impact articles within the corpus.

The existence of a high number of unconnected nodes indicates that the domain has a significant gap in the literature reviews, which helps bring the domain together by analyzing and comparing concepts in different articles against each other.

7.2. RQ14: Which Countries Are Involved in Research in This Domain?

Several countries are involved in research on affective computing in education. Exactly 44 countries have authored the selected articles. Figure 16 describes the citation network of these countries. The nodes of the graph represent the countries authoring the reviewed articles. The edges represent citation links between two countries, with articles from one citing articles from the other. The size of a node is proportional to the sum of citations of the articles authored by the corresponding country. The link strength, i.e., the thickness of an edge, is proportional to the number of citations between the corresponding countries.

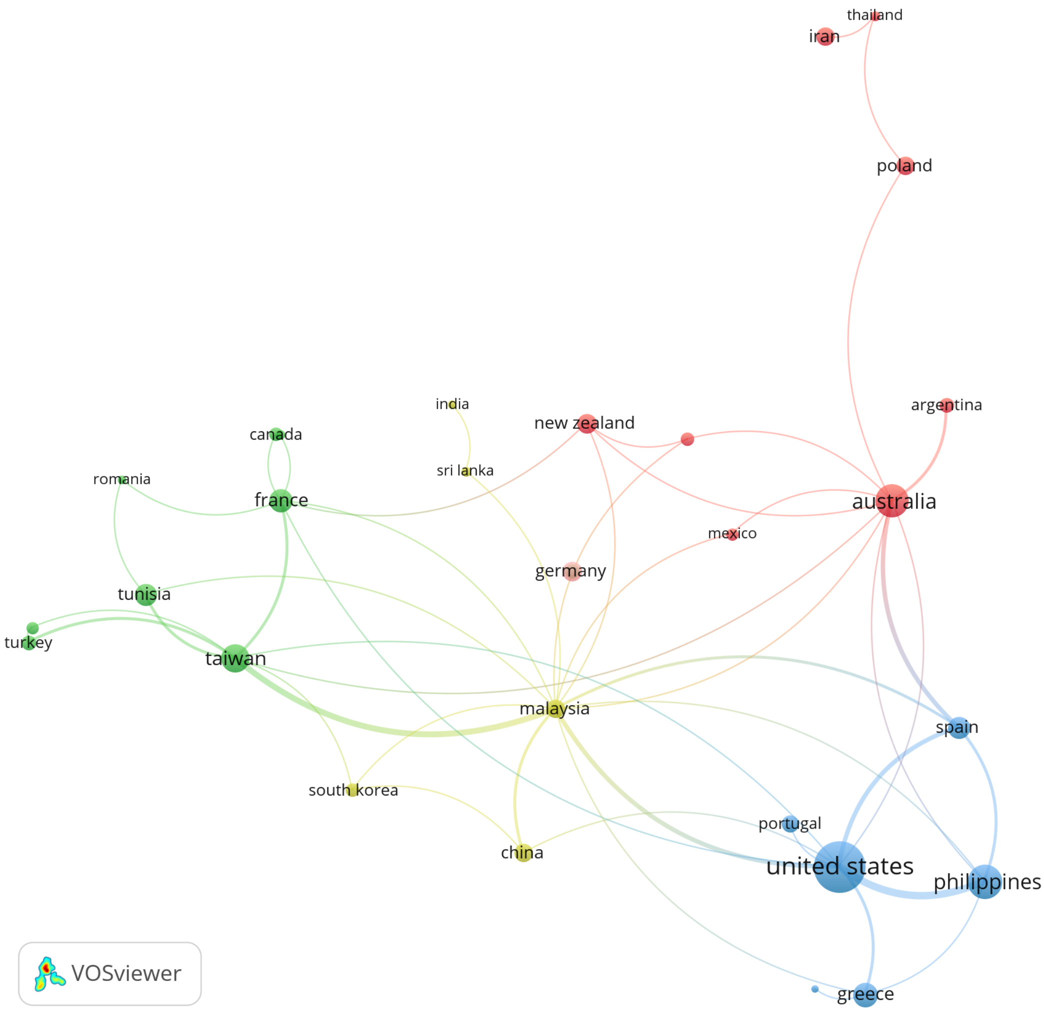

Figure 16. Citation links of authoring countries of reviewed articles. The normalization method used is the association strength normalization method. The resolution is 3.00, and the minimum cluster size is 2.

Table 6 is a list of the 11 most cited countries and the number of selected articles authored by them. The United States has the highest number of citations at 1651, followed by the Philippines and Australia. Incidentally, China has the second-highest number of articles at 17 while it only ranks eleventh in citations, which could result from a number of factors: highly specific research output from the country, poor quality of research, lack of exposure, and limited dissemination/visibility. Malaysia has links to 14 countries, the highest, indicating high international exposure of Malaysian researchers in the domain.

Table 6. Most cited authoring countries of reviewed articles.

Country	Citations	Articles
United States	1651	22
Philippines	455	2
Australia	411	8
Taiwan	234	9
Greece	147	4
France	128	4
Tunisia	111	4
Spain	97	8
New Zealand	78	2
Germany	65	3
China	63	17

7.3. RQ15: What Are the Current Research Directions in This Domain?

A total of 3697 articles have been referenced collectively by the selected articles. Figure 17 describes the co-citation network of these references. The nodes of the graph represent the articles referenced by the selected articles. The edges represent co-citation links between the two articles. Co-citation is a relation that links two articles that have been cited together by the same selected article. The size of a node is proportional to the total number of citations of the corresponding reference. The link strength, i.e., the thickness of an edge, is proportional to the number of co-citations between the corresponding references.

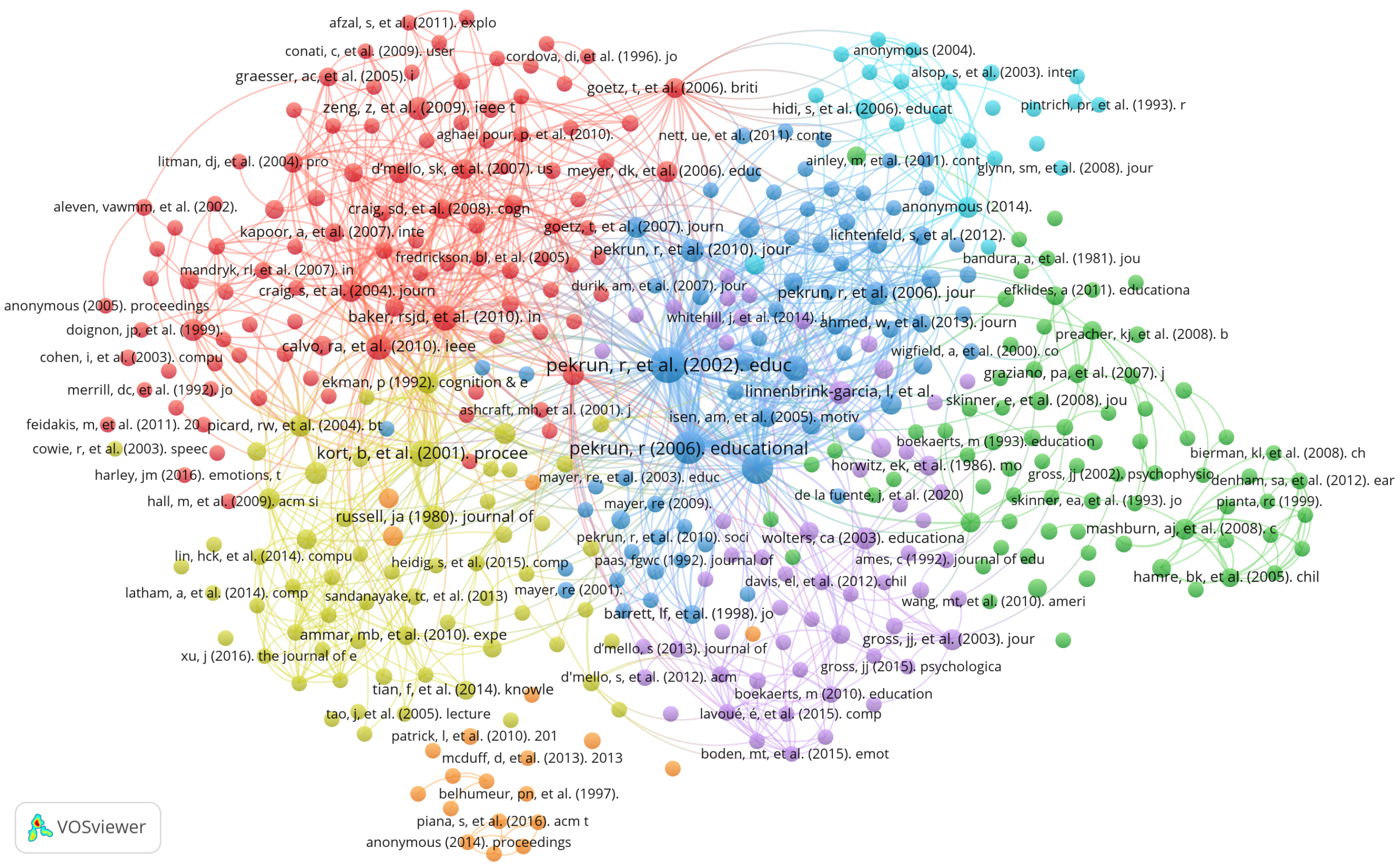

Figure 17. Co-citation links of the references in the corpus. The normalization method used is the association strength normalization method, with a resolution = 3.00 and minimum cluster size = 2.

It is important to note that some articles located toward the center of the clusters exhibit a high number of co-citation links, reflecting their prominence and continued relevance in the field. These articles tend to be older, having had more time to accumulate citations. For example, Baker et al. (2010), Kort et al. (2001) and Pekrun (2006), which are positioned centrally in the graph, demonstrate the sustained influence of foundational works in this area. The large number of co-citation links associated with these articles underscores their ongoing importance in shaping research on affective computing in education.

Analysis of the co-citation network of the references of the selected articles helps identify the tightly related literature that serves as the foundation of the contributions of the selected articles. Clusters of references that are cited together indicate a particular branch within the domain. Through Figure 17, it can be seen that VOSviewer identifies 7 clusters within the references, which are marked with different colors.

1. The largest cluster containing 104 references is marked red. These nodes correspond to referenced articles that design and analyze automated emotion recognition and regulation tools for learners. Craig et al. (2004) proposed AutoTutor, an ITS with natural language tutor dialogue. Baker et al. (2010) analyzed three different computer-based learning environments. D'Mello et al. (2008) analyzed an emotion recognition system with data collected using AutoTutor. Conati (2002) analyzed learners' emotions in educational games. Calvo and D'Mello (2010) reviewed the literature on affect detection methods and applications.
2. The next largest cluster (marked green) contains 77 references. These articles analyze children's affect, cognition, and emotional regulation in classroom environments. Mashburn et al. (2008) studied classroom quality and its effect on pre-kindergarten children's language and social skills. Similarly, Graziano et al. (2007) and Blair (2002) studied the factors influencing the emotions and cognition of school children. Fredricks et al. (2004) analyzed and proposed the importance of factoring student engagement in schools.
3. The third-largest cluster containing 72 references is marked dark blue. This is the cluster of references that propose basic theoretical models of emotion also specialized to different environments. Much of the foundational research on affective computing has been reported by Reinhard Pekrun in Pekrun et al. (2002), who proposed the theory on academic emotions; Pekrun (2006) presented the control-value theory of emotions; and Pekrun et al. (2011) proposed the achievement emotions questionnaire (AEQ). Similarly, Goetz et al. (2007) analyzed the relationships between students' academic emotions and flows from one to another.
4. The cluster marked yellow comprises 61 references. These references designed and analyzed ITSs. Kort et al. (2001) described the components of a "learning companion". An agent-based affective tutoring system has been developed (Mao & Li, 2010). Ben Ammar et al. (2010) proposed an affective tutoring system integrated within an ITS.
5. The fourth-largest cluster is marked purple. The 59 references in this cluster study the emotional self-regulation of learners. Wolters (2003) studied procrastination within self-regulated learning environments. Classroom motivation and emotions were analyzed by Boekaerts (2010). Gross and John (2003) measured the emotional regulation abilities of learners through the emotion regulation questionnaire (ERQ).
6. The final significant cluster is marked light blue and comprises 23 articles. These articles study the effect of classroom interventions on emotions and performance of learners. For example, Tomas Engel and Ritchie (2012) analyzed the effect of a hybridized writing activity on learners' emotions. Emotional energy within science demonstrations was studied by Milne and Otieno (2007). Similarly, Heddy and Sinatra (2013) used the teaching for transformative experiences in science (TTES) model of teaching and analyzed its effect on learners' emotions.

7. This cluster is composed of scattered references not connected to any major cluster. Many of these references propose datasets with emotion annotations for use in affective computing research. For example, Koelstra et al. (2012) presented a popular database for emotion analysis using physiological signals (DEAP). A mini-cluster exists within this group, which deals with emotion recognition through physical body gestures (Gunes & Piccardi, 2009; Piana et al., 2016).

While the co-citation network gives essential information about the research directions in the domain, the information is incomplete, because the selected articles are not included in the co-citation network. Hence, we also use a bibliometric coupling network of the selected articles to identify the exact current research trends in the domain.

Figure 18 describes the bibliometric coupling network of the selected articles. The nodes of the graph represent the selected articles. The edges represent bibliometric coupling links between two articles. Bibliometric coupling is a relation that links two articles that share the same references. The size of a node is proportional to the total number of citations of the referenced articles by the selected articles. The link strength, i.e., the thickness of an edge, is proportional to the number of co-citations between the corresponding references.

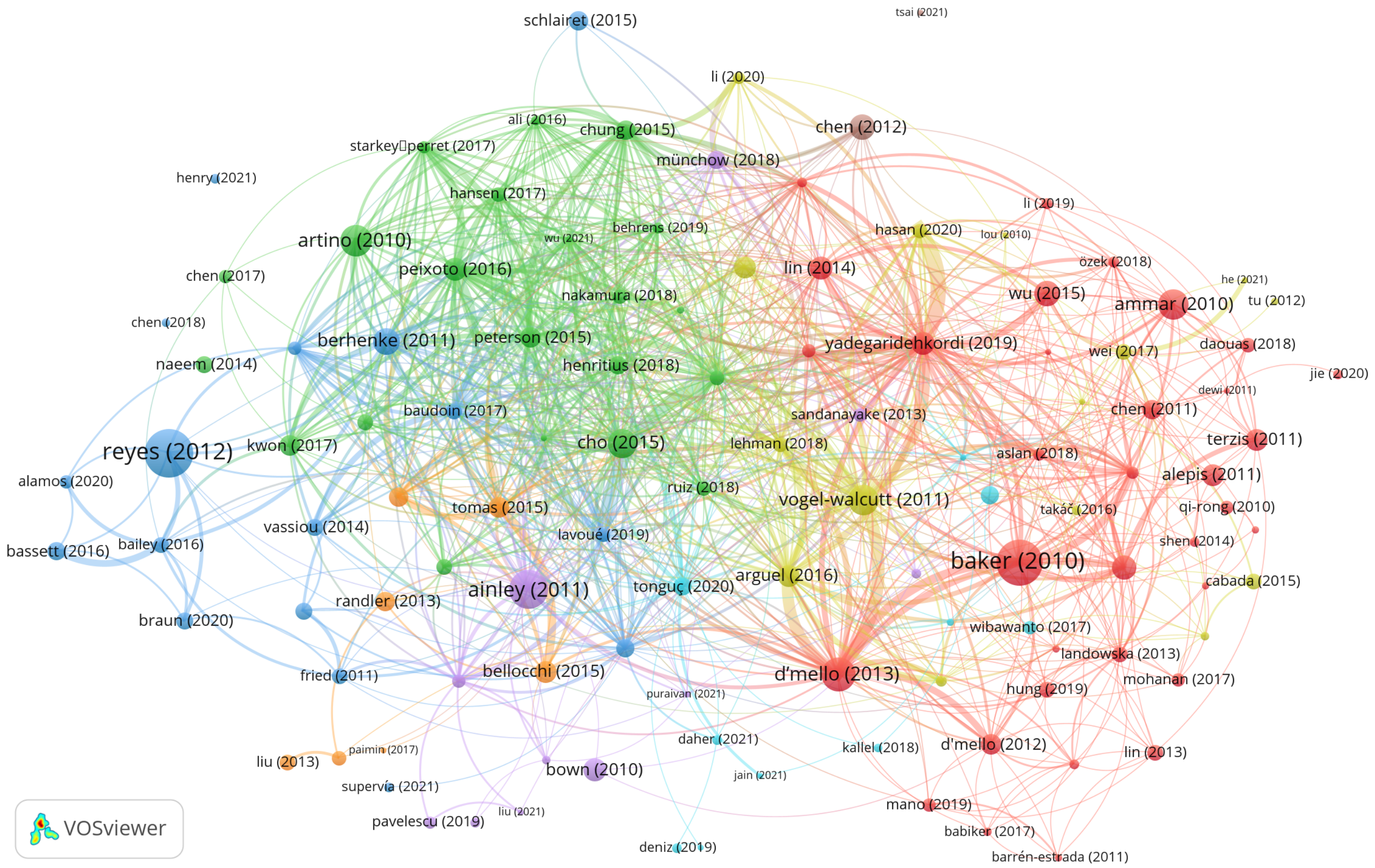

Figure 18. Bibliometric coupling links of reviewed articles. The normalization method used is the association strength normalization method, with a resolution = 3.00 and minimum cluster size = 2.

Bibliometric coupling is an important indicator of the relatedness of the two articles. Bibliometric coupling link strength is directly proportional to their relatedness. In Figure 18, VOSviewer identifies several clusters:

1. The largest cluster is marked red, and it consists of articles studying the emotions of learners with computer-assisted learning platforms. For example, Baker et al. (2010) studied the impact of CBLEs on the affective-cognitive states of learners. Similarly,

D'Mello (2013) measured the incidence of discrete affective states during technology-assisted learning. Terzis et al. (2013) measured the instantaneous emotions of learners during computer-based assessments using facial expressions.

2. The next largest cluster comprising 21 articles is marked in green. These articles studied learning emotions, especially academic achievement-related emotions. Starkey-Perret et al. (2017) studied the impact of teaching approaches on achievement emotions. Similarly, Peterson et al. (2015) studied achievement emotions during assessments. On the other hand, Peixoto et al. (2017) studied the relationship between achievement emotions and academic outcomes.
3. The cluster marked dark blue contains 17 articles that study students' emotional regulation within school/classroom environments. Bailey et al. (2015) studied preschooler emotional regulation abilities. Reyes et al. (2012), which was the most cited article, and Baudoin and Galand (2017), studied the effects of classroom environment and structure on students; emotions and outcomes; Braun et al. (2020) analyzed the effect of teachers' emotional regulation on students' emotions, and Lavoué et al. (2019) proposed tools to support emotional regulation in academics.
4. The 15 articles in the cluster marked in yellow proposed and analyzed ITSs. Myers (2021), who authored one of the latest articles within the corpus, proposed a methodology for the automatic detection of emotions for ITSs like AutoTutor. Bustos Lopez et al. (2021) proposed an academic resource recommendation system based on emotions. Similarly, Zatarain-Cabada et al. (2013) proposed an integration of learning styles and emotions within ITSs. Hasan et al. (2020) reviewed the transition from ITSs to ATSs.
5. There are some small clusters, the most notable of which are those marked orange. These articles study the role of affect in the learning domain of science and technology. For example, C.-J. Liu et al. (2013) reported students' mental states while learning about acids and bases in school. Similarly, Randler et al. (2012) described the specific emotion 'disgust' and its effect on the achievement of biology students. Tomas Engel et al. (2015) analyze students' emotional regulation in science classrooms.

From the clusters in Figures 17 and 18, it can be seen that while the overall research has been moving in directions like automated emotion recognition, theoretical models of emotion for different scenarios and metrics, and the impact of basic interventions on emotional regulation (as described in the co-citation network); recent research has been moving to more specific applications like ITSs, scientific learning domains, emotions in computer-assisted learning platforms, achievement emotions, etc.

8. Discussion

This section discusses some significant research opportunities and challenges identified by affective computing studies in the education domain, as required by the PRISMA 2020 guidelines.

8.1. Emotion Detection Versus the Need for Regulation Systems

As shown in RQ4, most of the research focuses on the design of systems that detect emotions, as shown by (Leong, 2020), or on building theoretical models that explore the influences of affect, cognition, achievement, and other factors on each other, as discussed by (Alazemi et al., 2023). As shown in RQ6, academic emotion detection systems have achieved a high accuracy. However, for these systems to be of any utility, they have to be integrated into wider systems, processes, and methodologies that can use that data to regulate the emotions of learners. Most of the proposed systems indulge in what the authors call 'passive' regulation: the system simply informs the tutors or the learners about the learning emotions, expecting the learners and tutors to cope accordingly. However, these systems are highly insufficient to be of much practical use. Most of the validation performed so far has been in the form of system usability studies that are highly empirical in nature and do not describe a quantified impact on achievement.

For there to be a true impact on educational achievement, 'active' academic emotion regulation systems need to be developed. These are systems that are integrated with an intelligent learning environment that can change course structure, material, pace, or digital tutor behavior based on the emotions detected by the detection system to better regulate the academic emotions of learners toward targeted levels of achievement. Some research is being done in this direction by Lei et al. (2022). However, these systems need to be tested outside the four walls on a larger scale. The following points of discussion also highlight the obstacles that we have identified through the review, toward a truly effective affective computing system for education.

8.2. Capturing Affective Data in Real Environments

Most systems with automated real-time affective data capturing components are limited to laboratories that are less complicated and have lower background noise (Tao & Tan, 2005). The existing data are primarily used for information retrieval and common feature identification, making it challenging to develop systems that effectively address complex changes in affect. Apart from developing high-quality emotional interaction, research should emphasize (i) building automatic affective information-capturing systems for real environments and (ii) obtaining more reliable and complex features.

8.3. Multi-Modal Affective Information Processing

Multi-modal data for measuring emotional responses offer more information for emotion recognition. Existing multi-modal approaches (Banda & Robinson, 2011; D'Mello & Kory, 2012) allow us to move around the limitations of individual measurement channels and enhance the accuracy compared to the best single-modal systems. These approaches leverage diverse modalities, such as facial expressions, physiological signals, voice tone, and text analysis, to capture the complex and multi-faceted nature of human emotions.

Nevertheless, several theoretical, methodological, and measurement challenges need to be addressed before employing these channels to assess learners' emotions experimentally. Theoretical challenges include understanding how emotions are represented across different modalities and whether combining them provides a unified or fragmented perspective of emotional states. This also involves identifying whether certain modalities contribute more significantly to specific emotional states or if their contributions vary across different contexts, such as individual or group learning settings.

Methodological challenges arise in designing experiments that can effectively integrate and validate multi-modal systems. For instance, achieving synchronization between modalities is critical, as physiological signals (e.g., heart rate variability) may lag behind facial expressions or verbal cues in capturing an emotional response. Researchers also face difficulties in determining the appropriate temporal granularity for data collection and analysis, ensuring the system remains sensitive enough to capture dynamic changes in emotional states while avoiding information overload.

Measurement challenges are particularly significant, as different modalities often provide conflicting or incomplete information about an emotional state. For example, facial expressions may indicate happiness while physiological signals suggest stress, leading to potential ambiguities in emotional classification. The integration of heterogeneous data types—such as continuous signals from biosensors and categorical self-reports—further complicates this process. Moreover, ensuring the reliability and validity of data collected across diverse modalities is a persistent issue, particularly when working with large, diverse populations or in ecologically valid settings.

Another major consideration is the scalability and adaptability of multi-modal systems. Real-world applications, such as educational technologies or adaptive learning platforms, require emotion recognition systems that can operate efficiently in real-time and across varying environments. This necessitates the development of computationally efficient algorithms for data fusion and interpretation. It also raises privacy concerns, as some modalities (e.g., facial recognition) may be perceived as intrusive by learners, affecting their behavior and the authenticity of emotional responses.

Future research must address these challenges by developing robust frameworks for multi-modal emotion recognition. This includes reconciling conflicts between modalities through advanced data fusion techniques, standardizing protocols for multi-modal data collection, and creating algorithms capable of interpreting complex emotional dynamics.

8.4. Scalability of Channel for Affective Measurement

The current state-of-the-art systems face significant challenges in scalability when applied to a wide range of learners. A major bottleneck lies in the inherent limitations of certain channels for affective measurement, particularly those that rely on specialized hardware or techniques. For example, while facial expression analysis is currently the most scalable channel due to its reliance on widely available cameras, it is not a comprehensive solution. Human emotions are complex and multi-faceted, expressed through a combination of facial expressions, vocal intonations, physiological signals, and behavioral cues, making reliance on a single channel insufficient. This highlights the need for multi-modal solutions that combine various channels, which are often constrained by technological, logistical, and practical barriers to scaling.

Intrinsic barriers to scalability in existing channels include the complexity of data collection and processing for physiological signals, such as EEG, ECG, or skin conductance. These channels require specialized, often cumbersome, devices and controlled environments to ensure data accuracy, making them unsuitable for large-scale or everyday use. Systemic barriers also exist, such as the high costs associated with deploying these devices across large student populations and the significant computational power required for real-time processing and storage of large multi-modal datasets.

As wearable sensors become more popular (Ba & Hu, 2023), these systemic barriers may be alleviated to some extent through advancements in technology. For example, recent innovations, such as haptic feedback from students' writing styluses (Schrader & Kalyuga, 2020), demonstrate the potential for scalable, less intrusive channels for affective measurement. However, these technologies are still in their infancy and require further refinement to meet the demands of scalability and accuracy in diverse educational contexts.

Another critical barrier to scalability is the handling and analysis of large volumes of data generated by continuous affective computing systems. Multi-modal data, which combine multiple channels, often lead to data heterogeneity, creating challenges for integration and analysis. Current workflows for handling such data are not designed to scale seamlessly across different educational settings, ranging from small classrooms to large institutions. To address these challenges, robust cloud-based platforms capable of real-time processing and storage must be developed, alongside standardized APIs and middleware to ensure compatibility with existing educational technologies.

For affective computing to be adopted widely in education, these systems must also prioritize ecological validity and practicality. As reviewed in RQ6, systems relying on physiological measures, such as EEG data, are the most accurate but are limited by their intrusiveness and cost. For example, using 24-channel EEG headsets is impractical for daily classroom use. To overcome this, advancements in portable, non-intrusive physiological devices are needed, making them more learner-friendly and less disruptive to the educational experience.

Real-world applications of affective computing include personalized learning, where content is tailored to a student's emotional state, and enhanced engagement, where teachers receive real-time feedback on student engagement levels. To scale these applications effectively, institutions must adopt cloud-based solutions and collaborate with tech companies to create user-friendly systems. Pilot studies in smaller settings can help refine these tools before broader implementation, ensuring they are optimized for diverse educational environments.

8.5. Integration Challenges

Integrating affective computing systems with existing educational technologies and infrastructures presents several challenges, including technical integration, privacy and security, and implementation and adaptation. Technical integration is particularly complex in the education domain due to the need to combine highly diverse data types—such as continuous physiological signals, discrete behavioral metrics, and emotional assessments—into coherent and actionable insights. Unlike many other domains, educational systems must also ensure real-time responsiveness and minimize disruption to learning processes, requiring compatibility with a wide variety of educational platforms, tools, and data standards that vary significantly between institutions. For instance, integrating emotion recognition data into a Learning Management System (LMS) or adaptive learning platform necessitates seamless synchronization to provide immediate feedback without compromising system performance. By adopting industry standards and using APIs and middleware, compatibility and data integration can be enhanced, enabling these systems to function effectively within diverse educational environments. Privacy and security are critical concerns, given the sensitivity of affective data. This necessitates stringent measures for secure storage, transmission, and ethical data usage. Affective data often contain deeply personal information, such as stress or anxiety levels, which could potentially be misused if not handled responsibly. To tackle these concerns, robust encryption protocols and data anonymization techniques must be implemented, along with comprehensive data use policies that ensure transparency and ethical handling of data. Implementation and adoption hurdles include the need for extensive training for educators, resistance to new technologies, and the high costs associated with deploying affective computing systems. Educators may require professional development programs to understand how to interpret and act on the data provided by these systems effectively. Applying change management strategies to address resistance to technology adoption, coupled with seeking funding opportunities, can support the smooth integration of affective computing systems in educational settings.

8.6. Affective Dataset

As shown in RQ3, the lack of affective datasets has limited the progress of current studies in affective computing in education/learning. While there are many datasets for human emotions, there are very few datasets dealing specifically with the academic emotions of learners. For research in this field to grow, it is important to have datasets specifically oriented toward education/learning. Establishing a dataset with a significantly high amount of affective data, especially multi-modal affective data, is necessary for affective computing.

Very few of the reviewed articles provide their data publicly. Surprisingly, the four articles that claim to provide a public affective dataset do not, in fact, offer one. Data are crucial for researchers aiming to design newer, more accurate interpretations from existing theoretical models of emotion. Further research should be involved in the design, collection, marking, search, tool-making, and other procedures associated with effective data in the education domain.

8.7. Enhancing Learning with Affective Computing

Learning is not just cognitive but also social and emotional. Optimizations of learning need to consider how the social and emotional dimensions are enhanced in learning processes. There are at least two key research directions for the affective computing field to consider in the future.

First, while research on affective computing has made promising progress in emotion recognition, more attention should focus on emotion regulation to enhance the utility of affective computing in education. Findings in RQ4 suggest most emotion-aware reinforcement systems seek to shift learners' affective states by changing the tone, expression, or emotion of the tutoring agent. These are useful but limited attempts at emotion regulation. More is needed, for example, on how intelligent tutoring systems can recommend learning activities that engage learners both cognitively and emotionally. Engaging learners with more appropriate learning activities is a deeper and more meaningful way of regulating their emotions than simply using the tone, expression, and emotion of the tutoring agent.

Second, with collaborative learning being one of the essential forms of 21st-century learning, affective computing, particularly the emotion regulation interventions reviewed, should contribute to the optimization of collaborative learning. This line of research could further expand the emotion recognition and regulation research reviewed in RQ4. One example is emotional detection systems that simultaneously monitor the synchrony of emotional states of a group of learners when they learn in collaboration. Exploring the synchrony of a group of learners' cognitive and emotional engagements simultaneously gives rise to a better understanding of learning and provides feedback to teachers or classroom orchestration systems for intervening and optimizing collaborative learning. Initial results proposed by E. J. Yang and Jin (2022) look promising.

8.8. Ethical Considerations

The ethical implications of applying affective computing to education have not yet been explored thoroughly. Many ambiguities in the current context (including legal frameworks) need to be discussed before large-scale applications of affective computing in education can be deployed. Information about a person's affective state is one of their most private data. Naturally, with the development of more robust affective computing applications in education, the ethics behind data handling and applications are of increasing importance. While there is no significant uniqueness of the education domain in an ethical framework, the ethics related to the domain are mainly applicable to affective computing in general.

Ong (2021) presented an ethical framework for affect-aware artificial intelligence (AI) systems. Four stakeholders, in addition to the AI system, namely, developers, regulators, operators, and emoters are identified. In the education domain, the operators and emoters can be both students and teachers, depending on the specific system. Developers comprise the entity that devised and deployed the application, while the regulators can range from institutional management to national governments. Two notions are identified to ensure the ethical correctness of the affective computing system in education. 'Provable beneficence' states that the students should benefit from the affective system and that these benefits should outweigh the material and immaterial costs incurred by students. Concepts such as minimization of bias within the sample group, scientific verifiability, transparency of the system, and accountability within the system are explored within this notion. The notion of 'Responsible stewardship' deals with the role of the operator. The operator needs to ensure that the intended effect of the system matches the outcome, and privacy, consent, and ownership of data should be clearly defined and adhered to. Also, the operator has to ensure that the data are not misused for other purposes, i.e., its specific application should be clearly defined. For example, the operator should ensure that the system does not cause cognitive overload for the students (C.-H. Wu et al., 2015). Additionally, educators should ensure ethical considerations are in place before any student or learner data are collected when the affective system assistive tool is being used.

The use of affective computing in education raises several ethical implications, particularly concerning data privacy, consent, and the potential psychological impact on students. Data privacy is a paramount issue, as affective computing involves the collection and analysis of highly sensitive emotional and physiological data, necessitating stringent measures to protect this information from unauthorized access and misuse. Obtaining informed consent is critical; students and parents must be fully aware of what data are being collected, how it will be used, and the potential risks involved. Ensuring transparency and giving individuals the option to opt-out is essential for maintaining trust. Establishing clear consent protocols with comprehensive information and easy opt-out options ensures ethical transparency. To address these issues, implementing strong encryption and anonymization techniques can safeguard data privacy. Establishing clear consent protocols with comprehensive information and easy opt-out options ensures ethical transparency. To mitigate psychological impacts, schools should provide counseling support and use affective data cautiously, ensuring it is interpreted by trained professionals to avoid misjudgments. Creating an ethical framework that prioritizes student well-being, autonomy, and data security is essential for the responsible integration of affective computing in education.

8.9. More Research Gaps

This review revealed some major gaps in the current state of the research in affective computing in the education domain.

There is a big gap in research analyzing the emotional needs of special-needs students, disabled students, and students from historically marginalized classes. Similarly, the number of adults returning to college is on the rise. Only Zhao and Wang (2023) studied the emotions of Chinese ethnic minority students. The emotional needs of learners from minority backgrounds are significantly different from their peers. It is also ethically important to incorporate these features before a viable affect detection and regulation system can be deployed on a significant scale.

While some of the reviewed articles explored the addition of affective expression capabilities to virtual online tutoring agents, there is a lack of research on the effect of a physical automated tutor (for example, a robot) on students' emotions. It remains to be seen whether a physical manifestation of a tutoring agent would be more successful in positively regulating students' emotions compared to an on-screen video tutor, as presented by Guo et al. (2014).

Future research in affective computing for education should focus on concrete, actionable advancements, particularly in integrating emerging technologies such as the Internet of Things (IoT) and AI, to create adaptive learning environments that respond dynamically to students' emotional states in real-time. By employing wearable devices and classroom sensors, educators can monitor emotional and physiological responses, facilitating personalized feedback and instructional strategies tailored to individual needs.

Deploying AI algorithms for emotion recognition, using multi-modal inputs such as facial expressions, voice intonation, and physiological signals, can enhance accuracy in diverse educational settings. For instance, recent deep learning algorithms (e.g., Vision

Transformers) have set new benchmarks in computer vision tasks by providing higher accuracy and efficiency. Future research could tap into this technology for affective computing by improving the accuracy of emotion recognition systems while reducing computational costs, making it more feasible for real-time applications in educational settings. Implementing adaptive learning systems that use AI to personalize educational content based on real-time affective data will tailor learning experiences to individual needs. The fields of augmented reality (AR) and virtual reality (VR) also provide an opportunity to be integrated as Immersive Learning Environments, while being based on affective computing theory, to improve learner's achievements and satisfaction (Lampropoulos et al., 2024).

Establishing comprehensive data governance frameworks for ethical data collection and use, as well as ensuring privacy and informed consent, is essential. Longitudinal studies and pilot programs in varied educational environments will help assess the long-term impact and refine these technologies for broader adoption. Furthermore, longitudinal studies are needed to assess the long-term impacts of affective computing interventions on student engagement, mental health, and academic performance, providing empirical evidence to guide implementation.

Also, it has been observed from our corpus that there is no standardized metric for the quantitative and qualitative evaluation of affective computing interventions in education. This lack of standardization makes it challenging to compare the performance of different systems and interventions effectively. Researchers often rely on bespoke metrics tailored to their specific study contexts, which limits the generalizability of their findings. Furthermore, the absence of agreed-upon benchmarks hinders the ability to identify best practices and establish guidelines for designing and evaluating these systems. Addressing this gap by developing standardized evaluation frameworks could significantly advance the field, fostering consistency and enabling meaningful cross-study comparisons.

9. Limitations

While this systematic review and bibliometric analysis provide a comprehensive overview of affective computing in education from 2010 to 2023, several limitations should be acknowledged according to the PRISMA 2020 guidelines.

First, the review was limited to articles published in English. This language restriction may have excluded relevant studies published in other languages, potentially introducing a language bias and limiting the global perspective of the findings.

Second, although we conducted an extensive search across five major bibliographic databases—Science Citation Index Expanded (SCI), Social Science Citation Index (SSCI), IEEE Xplore, ACM Digital Library, and PubMed—it is possible that relevant studies indexed in other databases or unpublished studies (gray literature) were not included. This might affect the comprehensiveness of the review and could lead to publication bias.

Third, the search strategy, while carefully formulated, relied on specific keywords related to affective computing and education. There is a possibility that some relevant articles were missed due to variations in terminology or indexing practices, which is a common limitation in systematic reviews.

Fourth, the review focused specifically on affective computing applications that support learning in education. Studies that primarily addressed teaching practices, teachers' emotions, or administrative aspects of education were excluded. While this focus ensures depth in the selected area, it may limit the breadth of insights into the broader field of affective computing in education.

Fifth, the inclusion and exclusion criteria, as well as the study selection process, were conducted rigorously by two independent reviewers to minimize bias. However, the possibility of subjective judgment influencing the selection and data extraction cannot be entirely eliminated, which is a recognized limitation in systematic reviews.

Sixth, the heterogeneity of the studies included—in terms of research designs, methodologies, sample sizes, educational settings, channels for affective measurement, and theoretical frameworks—poses a challenge for generalizing the findings. This diversity makes it difficult to perform meta-analyses or to draw definitive conclusions applicable across different contexts.

Seventh, the lack of publicly available datasets and the limited reporting of methodological details in some studies restricted our ability to conduct a more in-depth analysis. The scarcity of open-access data also hinders replication studies and the advancement of research in this field.

Finally, the rapidly evolving nature of technology and educational practices means that findings may become outdated quickly. The cut-off date for our knowledge is October 2023, and subsequent developments in affective computing and education may not be reflected in this review.

These limitations highlight the need for future research to include multilingual studies, consider gray literature, employ broader search strategies, and promote transparency and standardization in reporting. Additionally, there is a need for more large-scale, longitudinal studies with diverse populations to enhance the generalizability of findings in affective computing for education.

10. Conclusions

In this article, we conducted a near-exhaustive systematic review and bibliometric analysis of literature related to affective computing in education, by following the PRISMA 2020 guidelines. Various aspects of the current state-of-the-art were studied and presented. The research results display that the number of publications has considerably increased in recent years. This systematic review explored affective computing applications in education, examining the research questions in the context of recent studies, tools, and methodologies. Below, each research question's findings provide a synthesized perspective on the state and implications of affective computing in educational environments.

RQ1: Learning domains: STEM fields, covering approximately 35% of studies, emerge as a significant focus within affective computing due to their cognitive demands, which often necessitate emotion regulation to improve engagement and performance. Second-language acquisition follows closely, emphasizing emotions like motivation and anxiety, essential to mastering new languages. Other fields, including medical and nursing education, are studied for their distinctive emotional dynamics, contributing valuable insights into affective needs across diverse educational domains.

RQ2: Affective measurements—researchers predominantly use facial expression analysis (30%) and self-reported questionnaires (40%) to measure students' emotional states. While questionnaires capture self-assessed emotions and engagement levels, facial analysis provides real-time emotion data, though often requiring controlled settings. Physiological data such as EEG (12%) and heart rate (10%) offer high precision, highlighting the balance between technological sophistication and ecological validity in affective computing.

RQ3: Datasets—a limited number of public datasets restrict broader research efforts in this field. Notably, only a few datasets, like FER2013 for facial expression and DEAP for physiological signals, are widely available, with most studies collecting proprietary data that remains inaccessible. The lack of standardized, shared datasets poses a barrier to progress, limiting opportunities for validation, cross-study comparison, and the development of universally applicable affective models.

RQ4: Types of affective systems—current systems are primarily focused on emotion recognition (65%) rather than emotion regulation. Most studies assess learners' emotional states passively and lack active interventions to adjust these emotions in real-time. Although some systems provide feedback to educators or learners, there is a gap in adaptive systems that dynamically alter learning experiences to enhance positive emotions and mitigate negative ones.

RQ5: Validation environments—research is conducted across online (30%), classroom (45%), and laboratory (25%) settings. Classroom environments provide realistic insights but often lack the precision of laboratory-controlled studies. Online studies are increasingly relevant due to the rise of digital learning platforms; however, they face challenges in accurately capturing emotions due to environmental variability.

RQ6: Intervention performance—the average accuracy across emotion detection systems is around 80%, with some high-precision models achieving up to 95%. While this accuracy is promising, it is predominantly achieved in controlled laboratory conditions. The performance of these systems in real-world educational settings is an area for further research to ensure robustness and generalizability.

RQ7: Theoretical models of emotion—the control-value theory of achievement emotions (CVTAE) framework is the most cited theoretical framework, emphasizing the role of control and value in emotional engagement. Other models like self-determination theory and social–cognitive theory offer valuable perspectives, though they are less frequently applied in educational settings, suggesting an opportunity to expand the theoretical grounding of affective computing.

RQ8: Reported emotions—studies frequently report basic emotions such as happiness, sadness, and anger, with learning-related states like boredom, frustration, and engagement also prominent. However, the lack of nuance in some emotion categories and the limited focus on positive learning emotions highlight a potential area for expanding affective metrics to enhance student well-being and engagement.

RQ9: Temporal trends—this field has seen steady growth, with publications doubling from 2017 to 2023. This trend suggests increased interest in affective computing, potentially driven by recent shifts to online learning and digital classroom tools that require new methods for understanding and enhancing student engagement.

RQ10: Publication venues—research is primarily published in high-impact journals like *Frontiers in Psychology* and *Computers in Education*. While journal publications suggest rigorous peer review, conference presentations (accounting for 30% of studies) provide avenues for sharing emerging work and innovations. Open-access articles (40%) are more widely cited, indicating the field benefits from accessible research.

RQ11: Research quality—the average quality score is moderate, around 5.4 out of 8, indicating reliable but variable rigor across studies. Articles often lack detailed intervention methods or data access, limiting reproducibility. Enhanced reporting standards and data-sharing practices could improve transparency and foster replication.

RQ12: Research purpose—nearly half of the studies focus on designing emotion recognition or expression systems, while fewer investigate the interplay between emotion and performance. This emphasis on detection over intervention highlights the need for systems that integrate real-time emotional data into adaptive learning tools, which could better serve educational outcomes.

RQ13: Bibliometric connections—articles often cite seminal works on automated emotion recognition and classroom emotion regulation, indicating foundational contributions in these areas. However, a large number of unconnected studies suggest fragmented research efforts, underscoring a need for integrative reviews and cross-study dialogue.

RQ14: Countries contributing—the United States leads in citations and publications, with strong contributions from China, the UK, and Australia. Increased cross-country collaborations could further diversify perspectives and strengthen the field, particularly in the context of educational systems with different cultural and structural characteristics.

RQ15: Research directions—the field is evolving from foundational emotion detection to specialized applications, such as intelligent tutoring systems (ITSs) and collaborative learning support. Future directions include adaptive learning systems, integration of physiological data for multi-modal emotion detection, and improved scalability for large classrooms. Emerging research is also looking into underserved populations, such as students with special needs, signaling a move toward more inclusive affective computing.

Based on the findings, this review aims to provide policymakers and practitioners in the education sector with the latest insights into the application of affective computing technology. However, it is important to acknowledge several critical limitations. The review was restricted to English-language articles and publications indexed in five major bibliographic databases, which may have excluded relevant studies and introduced language and publication biases. Additionally, the focus on affective computing applications in education, while ensuring depth, may have limited the scope of the findings by excluding studies on teaching practices and other aspects of educational settings. Furthermore, the heterogeneity of the included studies and the lack of publicly available datasets hinder the generalizability and reproducibility of the results. Despite these limitations, this review provides a valuable starting point for future research, highlighting the need for broader, multilingual studies, better data sharing, and more comprehensive theoretical models to advance the field of affective computing in education. A discussion in the form of a challenge highlighted the gaps and barriers in current research. All the data collected and generated in this article are available as part of a Zenodo repository described in the data availability statement.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/educsci15010065/s1, File S1: PRISMA 2020 Checklist; Table S1: Review data of reviewed articles.

Author Contributions: Conceptualization, R.Y., R.M., and A.A.P.; methodology, R.Y. and R.M.; validation, R.Y., R.M., and A.A.P.; investigation, R.Y., R.M., and A.A.P.; data curation, R.Y. and R.M.; writing—original draft preparation, R.Y. and R.M.; writing—review and editing, R.Y., R.M., A.A.P., and J.S.H.; funding acquisition, R.Y. and J.S.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Singapore Ministry of Education (MOE) through the Education Research Funding Program (ERFP) (grant no: OER 14/22 YR), which is overseen by NIE NTU, Singapore.

Institutional Review Board Statement: This is a review of previously published papers and, therefore, does not require approval from the university.

Informed Consent Statement: The present study is a systematic review of the literature, so it is not necessary to establish an Informed Consent Statement.

Data Availability Statement: The data associated with this Systematic Review are available in the Supplementary Materials. The entire corpus of data extracted from the systematic review of the included papers is provided in a publicly available Zenodo repository at the following link: https://doi.org/10.5281/zenodo.6970842 (accessed on 26 December 2024). The associated PRISMA 2020 checklist and figures from the bibliometric study are also available in the Supplementary Materials.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study, in the collection, analyses, or interpretation of data, in the writing of the manuscript, or in the decision to publish the results.

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